Applied Regression Analysis Using STATA

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Regression analysis is the statistical method most often used in social research. The reason is that most social researchers are interested in identifying ”causal” effects from non-experimental data. Regression is **the** method for doing this.

The term „Regression“: 1889 Sir Francis Galton investigated the relationship between body size of fathers and sons. Thereby he ”invented” regression analysis. He estimated

$$S_s = 85.7 + 0.56 S_F.$$

This means that the size of the son regresses towards the mean. Therefore, he named his method regression. Thus, the term regression stems from the first application of this method! In most later applications, however, there is no regression towards the mean.

1a) The Idea of a Regression

We consider two variables (Y, X). Data are realizations of these variables

$$(y_1, x_1), \ldots, (y_n, x_n)$$

resp.

$$(y_i, x_i), \qquad \text{for } i = 1, \ldots, n.$$

Y is the dependent variable, X is the independent variable (regression of Y on X). The *general idea of a regression* is to consider the conditional distribution

$$f(Y = y \mid X = x).$$

This is hard to interpret. The major function of statistical methods, namely to reduce the information of the data to a few numbers, is not fulfilled. Therefore one characterizes the conditional distribution by some of its aspects:

- Y metric: conditional arithmetic mean
- Y metric, ordinal: conditional quantile
- Y nominal: conditional frequencies (cross tabulation!)

Thus, we can formulate a regression model for every level of measurement of Y.

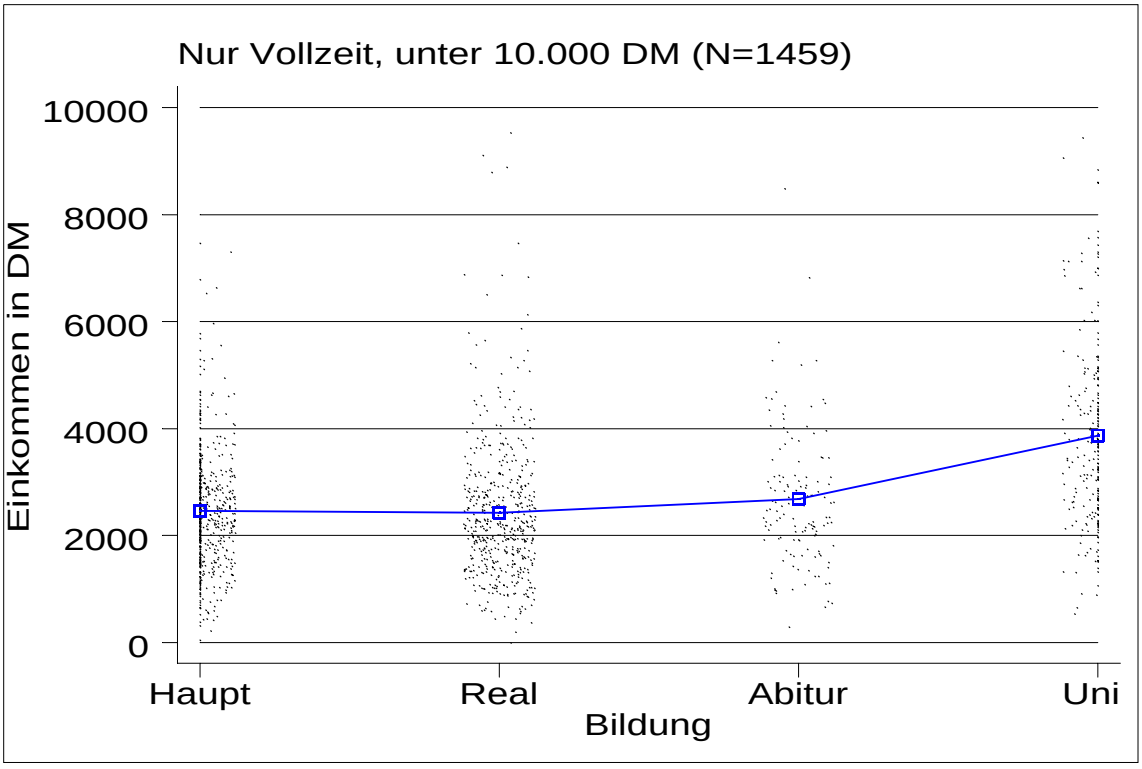
Regression with discrete X

In this case we compute for every X-value an index number of the conditional distribution.

Example: Income and Education (ALLBUS 1994)

Y is the monthly net income. X is highest educational level. Y is metric, so we compute conditional means $E(Y|x)$. Comparing these means tells us something about the effect of education on income (variance analysis).

The following graph is the scattergram of the data. Since education has only four values, income values would conceal each other. Therefore, values are ”jittered” for this graph. The conditional means are connected by a line to emphasize the pattern of relationship.



Regression with continuous X

Since X is continuous, we can not calculate conditional index numbers (too few cases per x-value). Two procedures are possible.

Nonparametric Regression

Naive nonparametric regression: Dissect the x-range in intervals (slices). Within each interval compute the conditional index number. Connect these numbers. The resulting nonparametric regression line is very crude for broad intervals. With finer intervals, however, one runs out of cases.

This problem grows exponentially more serious as the number of X's increases ("curse of dimensionality").

Local averaging: Calculate the index number in a neighborhood surrounding each x-value. Intuitively a window with constant bandwidth moves along the X-axis. Compute the conditional index number for every y-value within the window. Connect these numbers. With small bandwidth one gets a rough regression line.

More sophisticated versions of this method weight the observations within the window (locally weighted averaging).

Parametric Regression

One assumes that the conditional index numbers follow a function: $g(x;\theta)$. This is a parametric regression *model*. Given the data and the model, one estimates the parameters θ in such a way that a chosen criterion function is optimized.

Example: OLS-Regression

One assumes a linear model for the conditional means.

$$E(Y|x) = g(x;\alpha,\beta) = \alpha + \beta x.$$

The estimation criterion is usually "minimize the sum of squared residuals" (OLS)

$$\min_{\alpha,\beta} \sum_{i=1}^{n} (y_i - g(x_i;\alpha,\beta))^2.$$

It should be emphasized that this is only one of the many

possible models. One could easily conceive further models (quadratic, logarithmic, ...) and alternative estimation criteria (LAD, ML, ...). OLS is so much popular, because estimators are easily to compute and interpret.

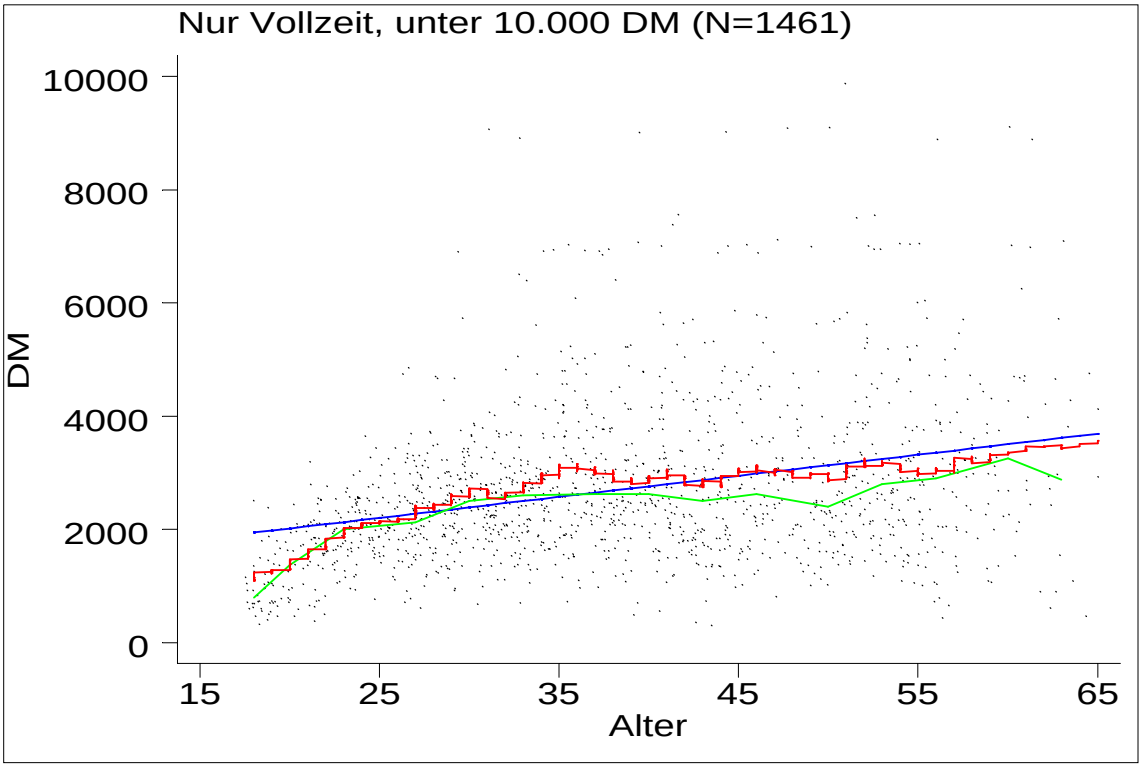
Comparing nonparametric and parametric regression

Data are from ALLBUS 1994. Y is monthly net income and X is age. We compare:

1) a local mean regression (red)

2) a (naive) local median regression (green)

3) an OLS-regression (blue)



All three regression lines tell us that average conditional income increases with age. Both local regressions show that there is non-linearity. Their advantage is that they fit the data better, because they do not assume an heroic model with only a few parameters. OLS on the other side has the advantage that it is much easier to interpret, because it reduces the information of the data very much ($\beta = 37.3$).

Interpretation of a regression

A regression shows us, whether conditional distributions differ for differing x-values. If they do there is an association between X and Y. In a multiple regression we can even partial out spurious and indirect effects. But whether this association is the result of a causal mechanism, a regression can not tell us. Therefore, in the following I do not use the term ”causal effect”. To establish causality one needs a theory that provides a mechanism which produces the association between X and Y (Goldthorpe (2000) On Sociology). Example: age and income.

1b) Exploratory Data Analysis

Before running a parametric regression, one should always examine the data.

Example: Anscombe's quartet

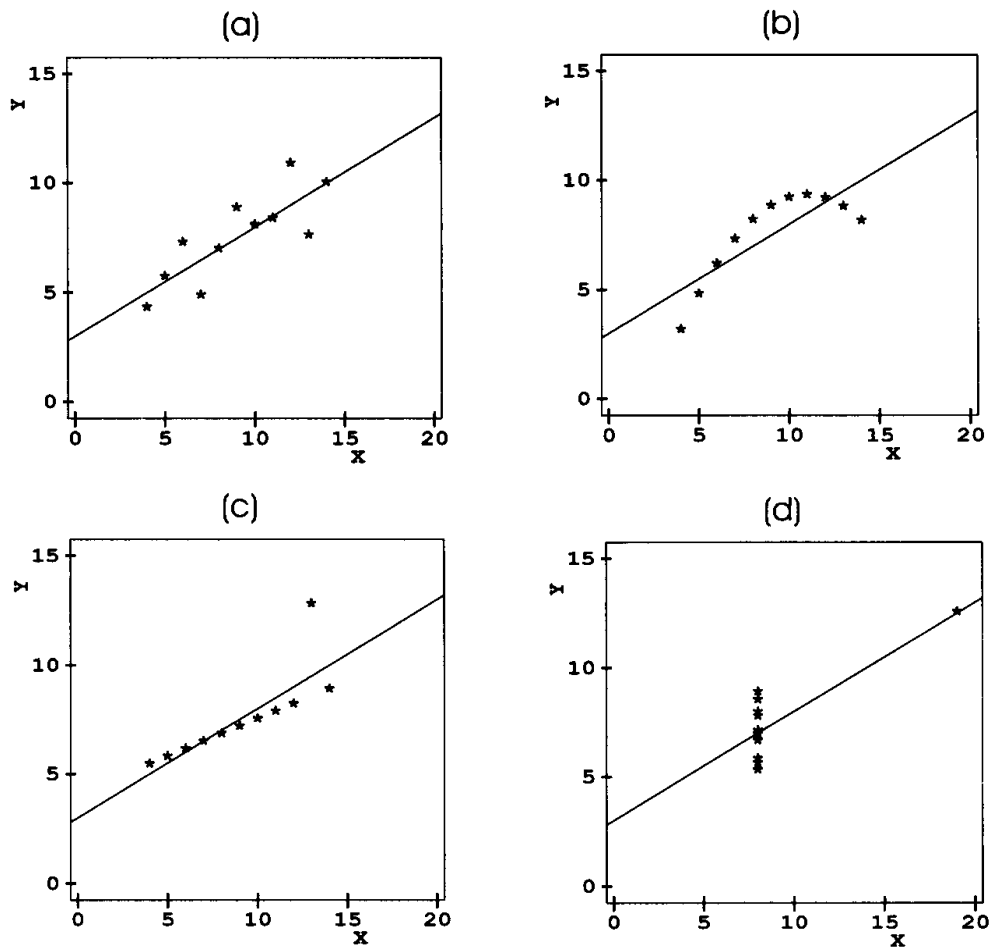
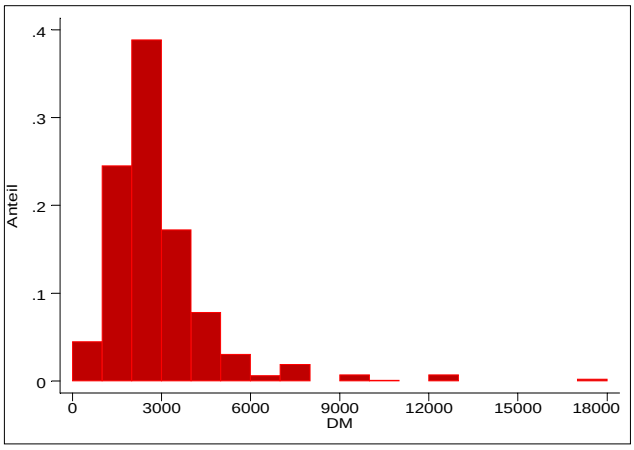


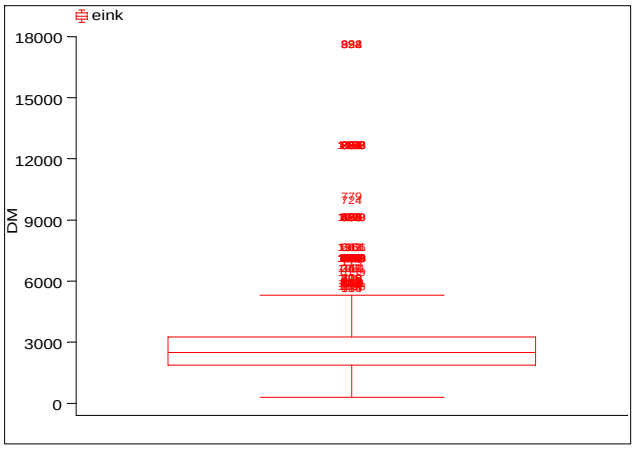
Figure 3.1. Four datasets, due to Anscombe (1973), with identical least-squares regressions. In (*a*), the linear regression is an accurate summary; in (*b*), the linear regression distorts the curvilinear relationship between Y and X; in (*c*), the linear regression is drawn toward an outlier; in (*d*), the linear regression "chases" the influential observation at the right.

Univariate distributions

Example: monthly net income (v423, ALLBUS 1994), only full-time (v25=1) under age 66 (v247≤65). N=1475.



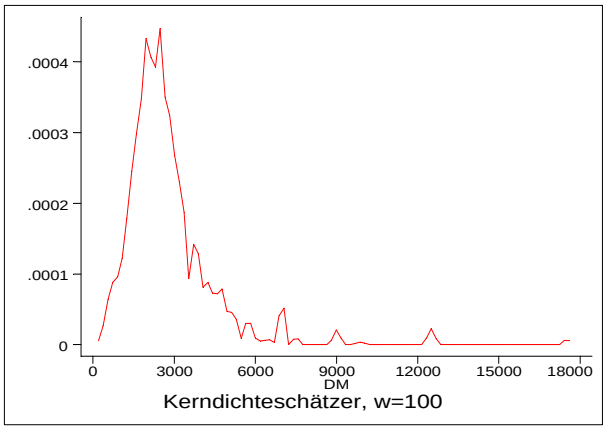
histogram

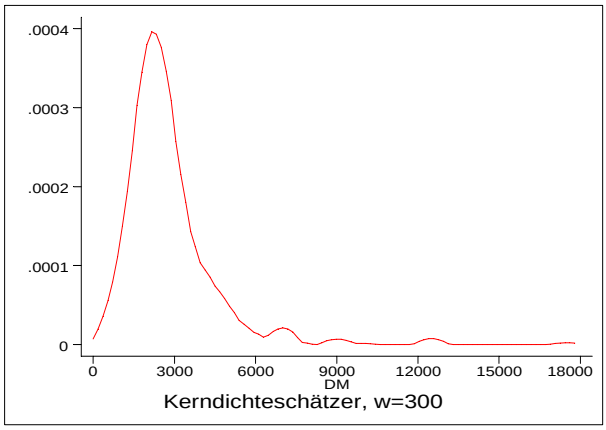


boxplot

The *histogram* is drawn with 18 bins. It is obvious that the distribution is positively skewed. The *boxplot* shows the three quartiles. The height of the box is the interquartile range (IQR), it represents the middle half of the data. The whiskers on each side of the box mark the last observation which is at most 1.5•IQR away. Outliers are marked by their case number. Boxplots are helpful to identify the skew of a distribution and possible outliers.

Nonparametric density curves are provided by the *kernel density estimator*. Density is estimated locally at n points. Observations within the interval of size 2w (w=half-width) are weighted by a kernel function. The following plots are based on an Epanechnikov kernel with n=100.

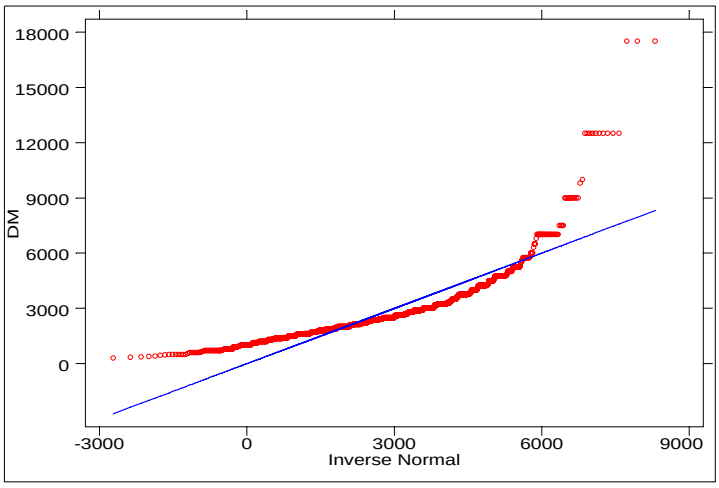




Comparing distributions

Often one wants to compare an empirical sample distribution with the normal distribution. A useful graphical method are *normal probability plots* (resp. normal quantile comparison plot). One plots empirical quantiles against normal quantiles. If the

data follow a normal distribution the quantile curve should be close to a line with slope one.



Our income distribution is obviously not normal. The quantile curve shows the pattern ”positive skew, high outliers”.

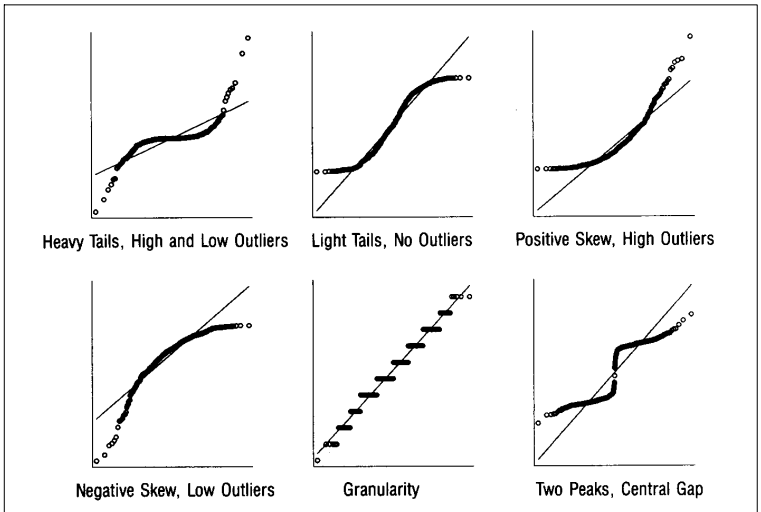


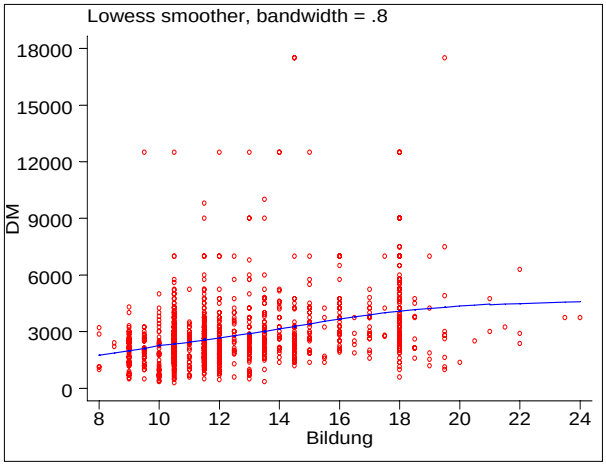
Figure 1.10 Quantile-normal plots reflect distribution shape.

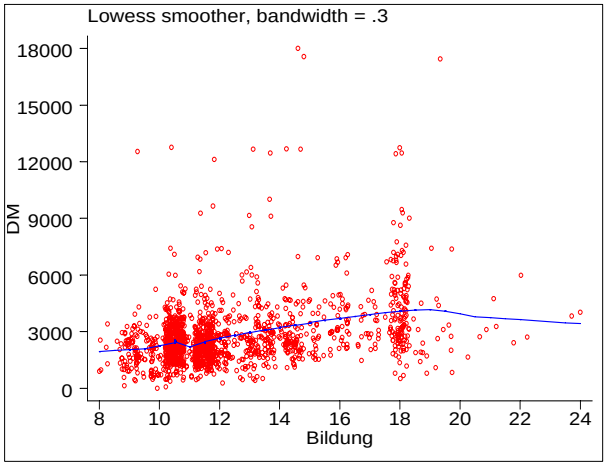
Bivariate data

Bivariate associations can best be judged with a *scatterplot*. The pattern of the relationship can be visualized by plotting a nonparametric regression curve. Most often used is the *lowess smoother* (locally weighted scatterplot smoother). One computes a linear regression at point x_i. Data in the neighborhood with a chosen bandwidth are weighted by a tricubic. Based on the estimated regression parameters $\hat{y}_i$ is computed. This is done for all x-values. Then connect $(x_i, \hat{y}_i)$ which gives the lowess curve. The higher the bandwidth is, the smoother is the lowess curve.

Example: income by education

Income defined as above. Education (in years) includes vocational training. N=1471.

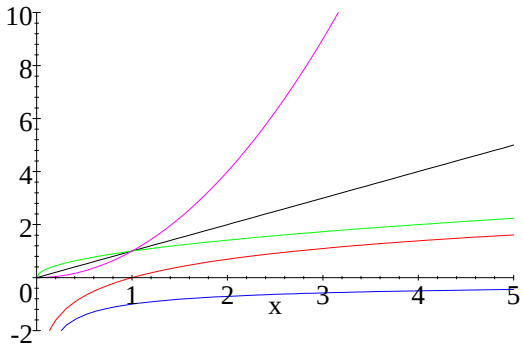




Since education is discrete, one should jitter (the graph on the left is not jittered, on the right the jitter is 2% of the plot area). Bandwidth is lower in the graph on the right (0.3, i.e. 30% of the cases are used to compute the regressions). Therefore the curve is closer to the data. But usually one would want a curve as on the left, because one is only interested in the rough pattern of the association. We observe a slight non-linearity above 19 years of education.

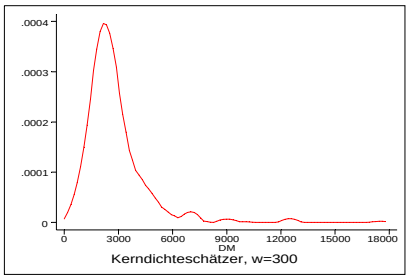
Transforming data

Skewness and outliers are a problem for mean regression models. Fortunately, power transformations help to reduce skewness and to ”bring in” outliers. Tukey’s ,,ladder of powers“:

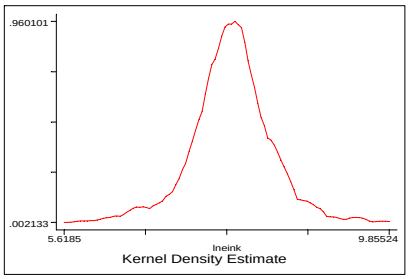


x^3	$q = 3$		apply if
$x^{1.5}$	$q = 1.5$	cyan	negative skew
x	$q = 1$	black	
$x^{.5}$	$q = .5$	green	apply if
$\ln x$	$q = 0$	red	positive skew
$-x^{-.5}$	$q = -.5$	blue	

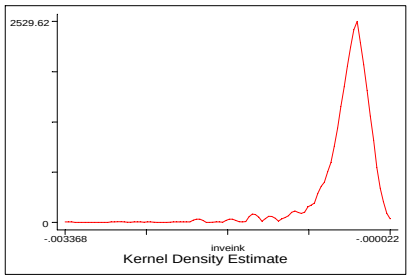
Example: income distribution



q=1



q=0



q=-1

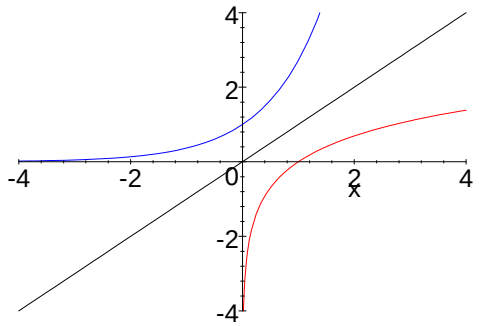
Appendix: power functions, ln- and e-function

$$x^{0.5} = x^{\frac{1}{2}} = \sqrt[2]{x}, \qquad x^{-0.5} = \frac{1}{x^{0.5}} = \frac{1}{\sqrt[2]{x}}, \qquad x^0 = 1$$

$\ln$ denotes the (natural) logarithm to the base $e = 2.71828\ldots$:

$$y = \ln x \Leftrightarrow e^y = x.$$

From this follows $\ln(e^y) = e^{\ln y} = y$.



some arithmetic rules

$e^x e^y = e^{x+y}$	$\ln(xy) = \ln x + \ln y$
$e^x / e^y = e^{x-y}$	$\ln(x/y) = \ln x - \ln y$
$(e^x)^y = e^{xy}$	$\ln x^y = y \ln x$

2) OLS Regression

As mentioned before OLS regression models the conditional means as a linear function:

$$E(Y|x) = \beta_0 + \beta_1 x.$$

This is the regression model! Better known is the equation that results from this to describe the data:

$$y_i = \beta_0 + \beta_1 x_i + \epsilon_i, \qquad i = 1, \ldots, n.$$

A parametric regression model models an index number from the conditional distributions. As such it needs no error term. However, the equation that describes the data in terms of the model needs one.

Multiple regression

The decisive enlargement is the introduction of additional independent variables:

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \ldots + \beta_p x_{ip} + \epsilon_i\,, \qquad i = 1, \ldots, n.$$

At first, this is only an enlargement of dimensionality: this equation defines a p-dimensional surface. But there is an important difference in interpretation: In simple regression the slope coefficient gives the *marginal* relationship. In multiple regression the slope coefficients are *partial* coefficients. That is, each slope represents the ”effect” on the dependent variable of a one-unit increase in the corresponding independent variable *holding constant* the value of the other independent variables. Partial regression coefficients give the direct effect of a variable that remains after controlling for the other variables.

Example: Status Attainment (Blau/Duncan 1967)

Dependent variable: monthly net income in DM. Independent variables: prestige father (magnitude prestige scale, values 20-190), education (years, 9-22). Sample: West-German men under 66, full-time employed.

First we look for the effect of status ascription (prestige father).

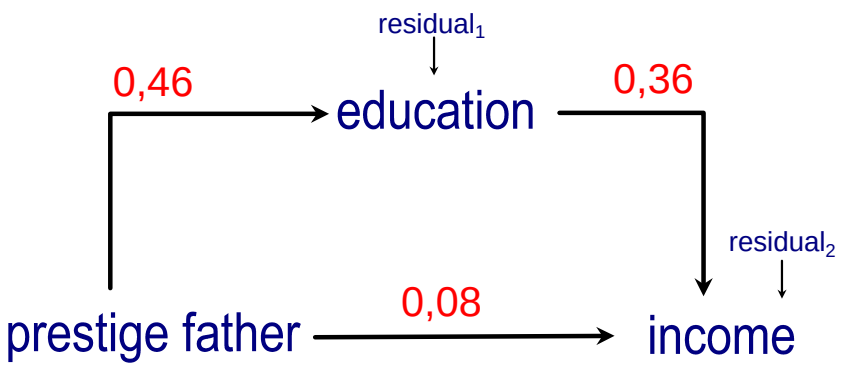
```
. regress income prestf, beta
```

```
Source |       SS       df       MS              Number of obs =     616
-------+------------------------------           F(  1,   614) =   40.50
 Model |   142723777     1   142723777           Prob > F      =  0.0000
Residu |  2.1636e+09   614  3523785.68           R-squared     =  0.0619
-------+------------------------------           Adj R-squared =  0.0604
 Total |  2.3063e+09   615  3750127.13           Root MSE      =  1877.2

------------------------------------------------------------------------
 income|      Coef.   Std. Err.      t    P>|t|                     Beta
-------+----------------------------------------------------------------
prestf |   16.16277   2.539641     6.36   0.000                  .248764
 _cons |   2587.704    163.915    15.79   0.000                        .
------------------------------------------------------------------------
```

Prestige father has a strong effect on the income of the son: 16 DM per prestige point. This is the marginal effect. Now we are looking for the intervening mechanisms. Attainment (education) might be one.

```
. regress income educ prestf, beta

Source |       SS       df       MS              Number of obs =     616
-------+------------------------------           F(  2,   613) =   60.99
 Model |   382767979     2   191383990           Prob > F      =  0.0000
Residu |  1.9236e+09   613  3137944.87           R-squared     =  0.1660
-------+------------------------------           Adj R-squared =  0.1632
 Total |  2.3063e+09   615  3750127.13           Root MSE      =  1771.4

------------------------------------------------------------------------
 income|      Coef.   Std. Err.      t    P>|t|                     Beta
-------+----------------------------------------------------------------
  educ |   262.3797   29.99903     8.75   0.000                 .3627207
prestf |   5.391151   2.694496     2.00   0.046                 .0829762
 _cons |  -34.14422   337.3229    -0.10   0.919                        .
------------------------------------------------------------------------
```

The effect becomes much smaller. A large part is explained via education. This can be visualized by a ”path diagram” (path coefficients are the standardized regression coefficients).



The direct effect of ”prestige father” is 0.08. But there is an additional large indirect effect 0.46•0.36=0.17. Direct plus

indirect effect give the total effect (”causal” effect).

A word of caution: The coefficients of the multiple regression are not ”causal effects”! To establish causality we would have to find mechanisms that explain, why ”prestige father” and ”education” have an effect on income.

Another word of caution: Do not automatically apply multiple regression. We are not always interested in partial effects. Sometimes we want to know the marginal effect. For instance, to answer public policy issues we would use marginal effects (e.g. in international comparisons). To provide an explanation we would try to isolate direct and indirect effects (disentangle the mechanisms).

Finally, a graphical view of our regression (not shown, graph too big):

Estimation

Using matrix notation these are the essential equations:

$$\mathbf{y} = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix}, \mathbf{X} = \begin{pmatrix} 1 & \mathrm{x}_{11} & \dots & \mathrm{x}_{1p} \\ 1 & \mathrm{x}_{21} & \dots & \mathrm{x}_{2p} \\ \vdots & \vdots & & \vdots \\ 1 & \mathrm{x}_{n1} & \dots & \mathrm{x}_{np} \end{pmatrix}, \boldsymbol{\beta} = \begin{pmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_p \end{pmatrix}, \boldsymbol{\epsilon} = \begin{pmatrix} \epsilon_1 \\ \epsilon_2 \\ \vdots \\ \epsilon_n \end{pmatrix}.$$

This is the multiple regression equation:

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}.$$

Assumptions:

$$\begin{aligned} \boldsymbol{\epsilon} &\sim N_n(\mathbf{0}, \sigma^2\mathbf{I}) \\ Cov(\mathbf{x}, \boldsymbol{\epsilon}) &= \mathbf{0} \quad . \\ rg(\mathbf{X}) &= p + 1 \end{aligned}$$

Estimation

Using OLS we obtain the estimator for $\boldsymbol{\beta}$,

$$\widehat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}.$$

Now we can estimate fitted values

$$\hat{\mathbf{y}} = \mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y} = \mathbf{H}\mathbf{y}.$$

The residuals are

$$\hat{\boldsymbol{\epsilon}} = \mathbf{y} - \hat{\mathbf{y}} = \mathbf{y} - \mathbf{H}\mathbf{y} = (\mathbf{I} - \mathbf{H})\mathbf{y}.$$

Residual variance is

$$\hat{\sigma}^2 = \frac{\hat{\boldsymbol{\epsilon}}'\hat{\boldsymbol{\epsilon}}}{n-p-1} = \frac{\mathbf{y}'\mathbf{y} - \mathbf{y}'\mathbf{X}\hat{\boldsymbol{\beta}}}{n-p-1}.$$

For tests we need sampling variances ($\hat{\sigma}_j$ standard errors are on the main diagonal of this matrix):

$$\hat{V}(\hat{\boldsymbol{\beta}}) = \hat{\sigma}^2(\mathbf{X}'\mathbf{X})^{-1}.$$

Squared multiple correlation is

$$R^2 = \frac{ESS}{TSS} = 1 - \frac{RSS}{TSS} = 1 - \frac{\sum \hat{\epsilon}_i^2}{\sum (y_i - \overline{y})^2} = 1 - \frac{\hat{\boldsymbol{\epsilon}}'\hat{\boldsymbol{\epsilon}}}{\mathbf{y}'\mathbf{y} - n\overline{y}^2}.$$

Categorical variables

Of great practical importance is the possibility to include categorical (nominal or ordinal) X-variables. The most popular way to do this is by coding dummy regressors.

Example: Regression on income

Dependent variable: monthly net income in DM. Independent variables: years education, prestige father, years labor market experience, sex, West/East, occupation. Sample: under 66, full-time employed.

The dichotomous variables are represented by one dummy. The polytomous variable is coded like this:

design matrix:

occupation	D1	D2	D3	D4
blue collar	1	0	0	0
white collar	0	1	0	0
civil servant	0	0	1	0
self-employed	0	0	0	1

One dummy has to be left out (otherwise there would be linear dependency amongst the regressors). This defines the reference group. We drop D1.

```
 Source |       SS       df       MS              Number of obs =    1240
---------+------------------------------           F(  8,  1231) =   78.61
   Model |  1.2007e+09     8   150092007           Prob > F      =  0.0000
Residual |  2.3503e+09  1231  1909268.78           R-squared     =  0.3381
---------+------------------------------           Adj R-squared =  0.3338
   Total |  3.5510e+09  1239  2866058.05           Root MSE      =  1381.8
\newpage
 ------------------------------------------------------------------------------
income |      Coef.   Std. Err.       t     P>|t|     [95% Conf. Interval]
 ------+-----------------------------------------------------------------------
educ   |   182.9042   17.45326     10.480   0.000    148.6628    217.1456
exp    |   26.71962   3.671445      7.278   0.000    19.51664     33.9226
prestf |   4.163393   1.423944      2.924   0.004    1.369768    6.957019
woman  |  -797.7655   92.52803     -8.622   0.000   -979.2956   -616.2354
east   |  -1059.817   86.80629    -12.209   0.000   -1230.122   -889.5123
white  |   379.9241   102.5203      3.706   0.000    178.7903     581.058
civil  |   419.7903   172.6672      2.431   0.015    81.03569    758.5449
self   |   1163.615   143.5888      8.104   0.000    881.9094    1445.321
_cons  |     52.905   217.8507      0.243   0.808   -374.4947    480.3047
 ------------------------------------------------------------------------------
```

The model represents parallel regression surfaces. One for each category of the categorical variables. The effects represent the distance of these surfaces.

The t-values test the difference to the reference group. This is not the test, whether occupation has a significant effect. To test this, one has to perform an incremental F-test.

```
. test white civil self

 ( 1)  white = 0.0
 ( 2)  civil = 0.0
 ( 3)  self  = 0.0

       F(  3,  1231) =   21.92
            Prob > F =    0.0000
```

Modeling Interactions

Two X-variables are said to interact when the partial effect of one depends on the value of the other. The most popular way to model this is by introducing a product regressor (multiplicative interaction). Rule: specify models including main and interaction effects.

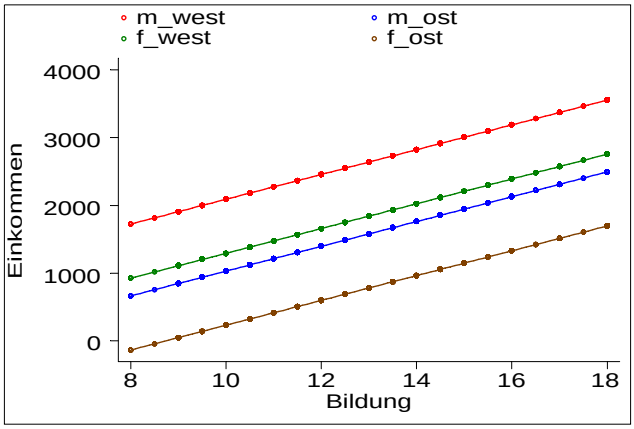
Dummy interaction

	woman	east	woman*east
man west	0	0	0
man east	0	1	0
woman west	1	0	0
woman east	1	1	1

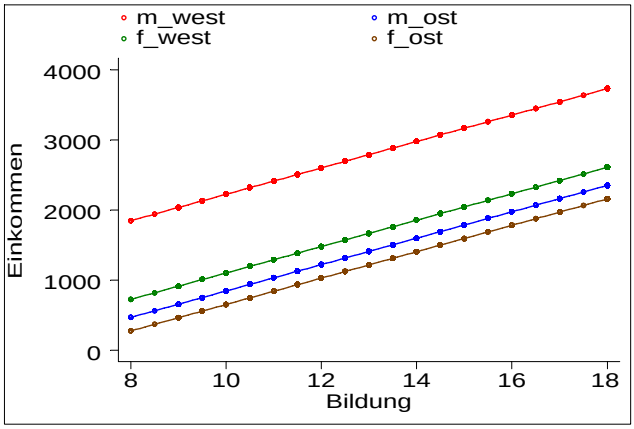
*Example: Regression on income + interaction woman*east*

```
 Source |       SS       df       MS              Number of obs =    1240
---------+------------------------------           F(  9,  1230) =   74.34
   Model |  1.2511e+09     9   139009841           Prob > F      =  0.0000
Residual |  2.3000e+09  1230  1869884.03           R-squared     =  0.3523
---------+------------------------------           Adj R-squared =  0.3476
   Total |  3.5510e+09  1239  2866058.05           Root MSE      =  1367.4

------------------------------------------------------------------------------
income |      Coef.   Std. Err.       t     P>|t|     [95% Conf. Interval]
-------+----------------------------------------------------------------------
educ   |   188.4242   17.30503     10.888   0.000    154.4736    222.3749
exp    |   24.64689   3.655269      6.743   0.000    17.47564    31.81815
prestf |    3.89539   1.410127      2.762   0.006     1.12887     6.66191
woman  |   -1123.29   110.9954    -10.120   0.000   -1341.051   -905.5285
east   |  -1380.968   105.8774    -13.043   0.000   -1588.689   -1173.248
white  |   361.5235   101.5193      3.561   0.000    162.3533    560.6937
civil  |   392.3995   170.9586      2.295   0.022    56.99687    727.8021
self   |   1134.405   142.2115      7.977   0.000    855.4014    1413.409
womeast|   930.7147    179.355      5.189   0.000    578.8392     1282.59
 _cons |   143.9125   216.3042      0.665   0.506   -280.4535    568.2786
------------------------------------------------------------------------------
```

Models with interaction effects are difficult to understand. Conditional effect plots help very much: exp=0, prestf=50, blue collar.



without interaction



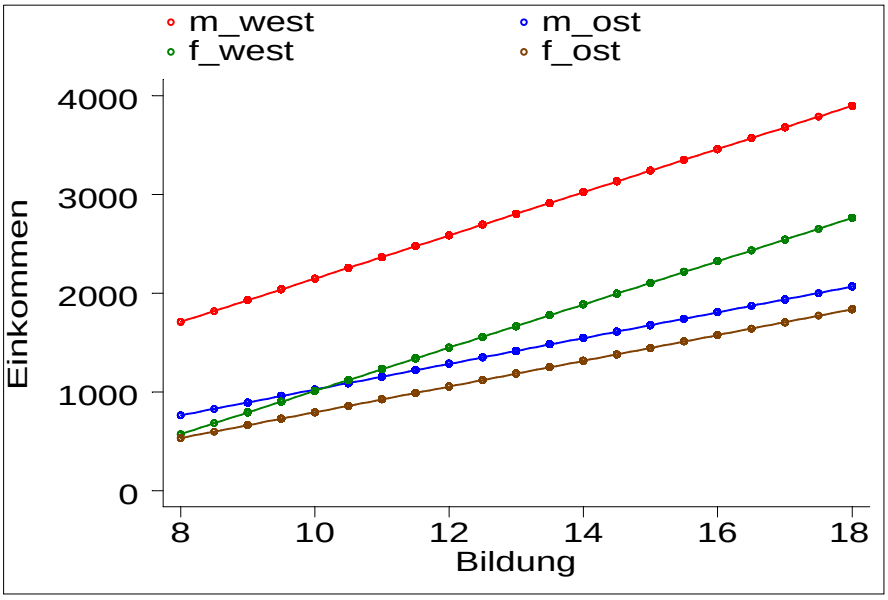
with interaction

Slope interaction

	woman	east	woman*east	educ	educ*east
man west	0	0	0	x	0
man east	0	1	0	x	x
woman west	1	0	0	x	0
woman east	1	1	1	x	x

*Example: Regression on income + interaction educ*east*

```
 Source |       SS       df       MS              Number of obs =    1240
---------+------------------------------           F( 10,  1229) =   68.17
   Model |  1.2670e+09    10   126695515           Prob > F      =  0.0000
Residual |  2.2841e+09  1229  1858495.34           R-squared     =  0.3568
---------+------------------------------           Adj R-squared =  0.3516
   Total |  3.5510e+09  1239  2866058.05           Root MSE      =  1363.3

------------------------------------------------------------------------------
income   |      Coef.   Std. Err.       t     P>|t|  [95% Conf. Interval]
---------+--------------------------------------------------------------------
educ     |   218.8579   20.15265     10.860   0.000  179.3205    258.3953
exp      |   24.74317    3.64427      6.790   0.000  17.59349    31.89285
prestf   |   3.651288   1.408306      2.593   0.010   .888338    6.414238
woman    |  -1136.907   110.7549    -10.265   0.000  1354.197   -919.6178
east     |  -239.3708   404.7151     -0.591   0.554  -1033.38    554.6381
white    |   382.5477   101.4652      3.770   0.000  183.4837    581.6118
civil    |   360.5762   170.7848      2.111   0.035  25.51422    695.6382
self     |   1145.624   141.8297      8.077   0.000  867.3686    1423.879
womeast  |   906.5249   178.9995      5.064   0.000  555.3465    1257.703
educeast |  -88.43585   30.26686     -2.922   0.004 -147.8163   -29.05542
   _cons |  -225.3985   249.9567     -0.902   0.367 -715.7875    264.9905
------------------------------------------------------------------------------
```



The interaction educ*east is significant. Obviously the returns to education are lower in East-Germany.

Note that the main effect of ”east” changed dramatically! It would be wrong to conclude that there is no significant income difference between West and East. The reason is that the main effect now represents the difference at educ=0. This is a consequence of dummy coding. Plotting conditional effect plots is the best way to avoid such erroneous conclusions. If one has interest in the West-East difference one could center educ ($educ - \overline{educ}$). Then the east-dummy gives the difference at the mean of educ. Or one could use ANCOVA coding (deviation coding plus centered metric variables, see Fox p. 194).

3) Regression Diagnostics

Assumptions do often not hold in applications. Parametric regression models use strong assumptions. Therefore, it is essential to test these assumptions.

Collinearity

Problem: Collinearity means that regressors are correlated. It is not a severe violation of regression assumptions (only in extreme cases). Under collinearity OLS estimates are consistent, but standard errors are increased (estimates are less precise). Thus, collinearity is mainly a problem of researchers who plug in many highly correlated items.

Diagnosis: Collinearity can be assessed by the variance inflation factors (VIF, the factor by which the sampling variance of an estimator is increased due to collinearity):

$$VIF = \frac{1}{1 - R_j^2},$$

where R_j^2 results from a regression of X_j on the other covariates. For instance, if R_j=0.9 (an extreme value!), then is $\sqrt{VIF}$=2.29. The S.E. doubles and the t-value is cut in halve. Thus, VIFs below 4 are usually no problem.

Remedy: Gather more data. Build an index.

Example: Regression on income (only West-Germans)

```
. regress income educ exp prestf woman white civil self
......
. vif
    Variable |       VIF       1/VIF
-------------+----------------------
       white |      1.65    0.606236
        educ |      1.49    0.672516
        self |      1.32    0.758856
       civil |      1.31    0.763223
      prestf |      1.26    0.795292
       woman |      1.16    0.865034
         exp |      1.12    0.896798
-------------+----------------------
    Mean VIF |      1.33
```

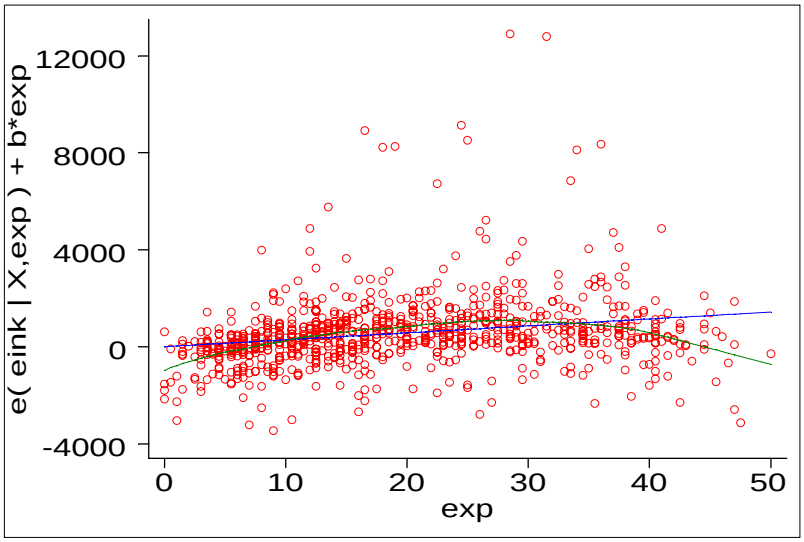
Nonlinearity

Problem: Nonlinearity biases the estimators.

Diagnosis: Nonlinearity can best be seen in the residual plot. An enhanced version is the component-plus-residual plot (cprplot). One adds $\hat{\beta}_j x_{ij}$ to the residual, i.e. one adds the (partial) regression line.

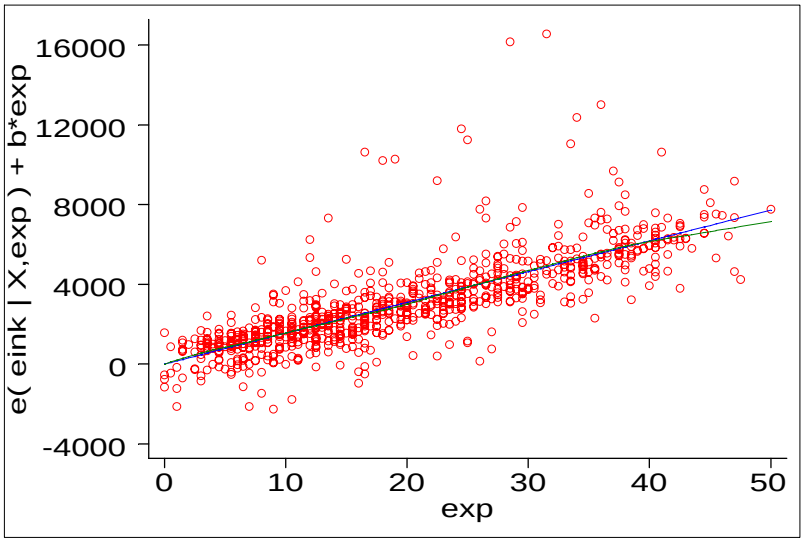
Remedy: Transformation. Using the ladder or adding a quadratic term.

Example: Regression on income (only West-Germans)



	β	t
Con	-293	
EXP	29	6.16
...		
N	849	
R^2	33.3	

blue: regression line, green: lowess. There is obvious nonlinearity. Therefore, we add EXP^2



	β	t
Con	-1257	
EXP	155	9.10
EXP^2	-2.8	7.69
...		
N	849	
R^2	37.7	

Now it works.

How can we interpret such a quadratic regression?

$$y_i = \beta_0 + \beta_1 x_i + \beta_2 x_i^2 + \epsilon_i, \qquad i = 1, \ldots, n.$$

Is $\widehat{\beta}_1 > 0$ and $\widehat{\beta}_2 < 0$, we have an inverse U-pattern. Is $\widehat{\beta}_1 < 0$ and $\widehat{\beta}_2 > 0$, we have an U-pattern. The maximum (minimum) is obtained at

$$X_{\max} = -\frac{\widehat{\beta}_1}{2\widehat{\beta}_2}.$$

In our example this is $-\frac{155}{2 \cdot -2.8} = 27.7$.

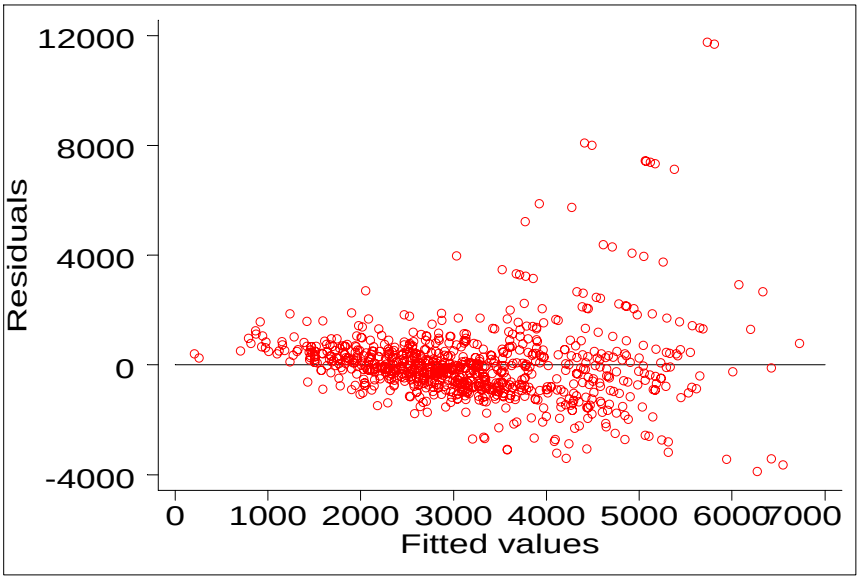
Heteroscedasticity

Problem: Under heteroscedasticity OLS estimators are unbiased and consistent, but no longer efficient, and the S.E. are biased.

Diagnosis: Plot $\widehat{\epsilon}$ against $\widehat{y}$ (residual-versus-fitted plot, rvfplot). Nonconstant spread means heteroscedasticity.

Remedy: Transformation (see below), WLS (one needs to know the weights, White-estimator (Stata option ”robust”)

Example: Regression on income (only West-Germans)



It is obvious that residual variance increases with $\widehat{y}$.

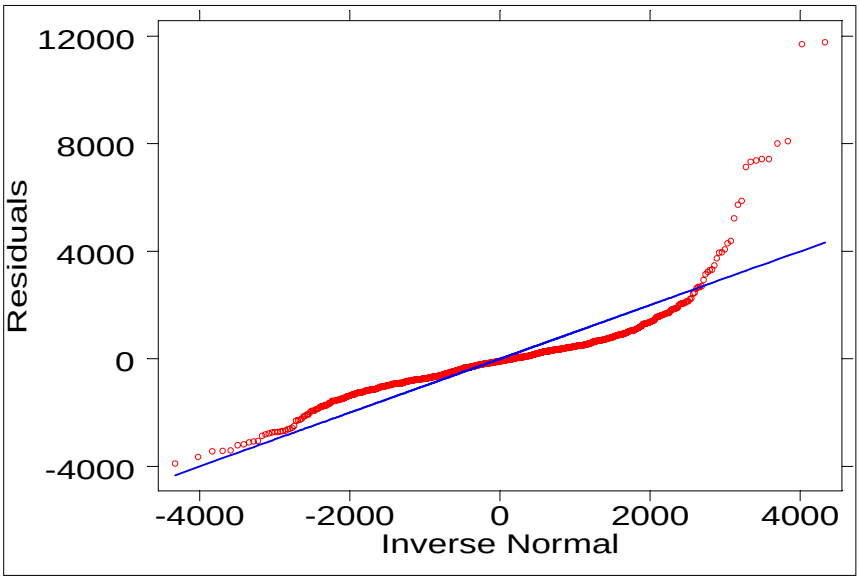
Nonnormality

Problem: Significance tests are invalid. However, the central-limit theorem assures that inferences are approximately valid in large samples.

Diagnosis: Normal-probability plot of residuals (not of the dependent variable!).

Remedy: Transformation

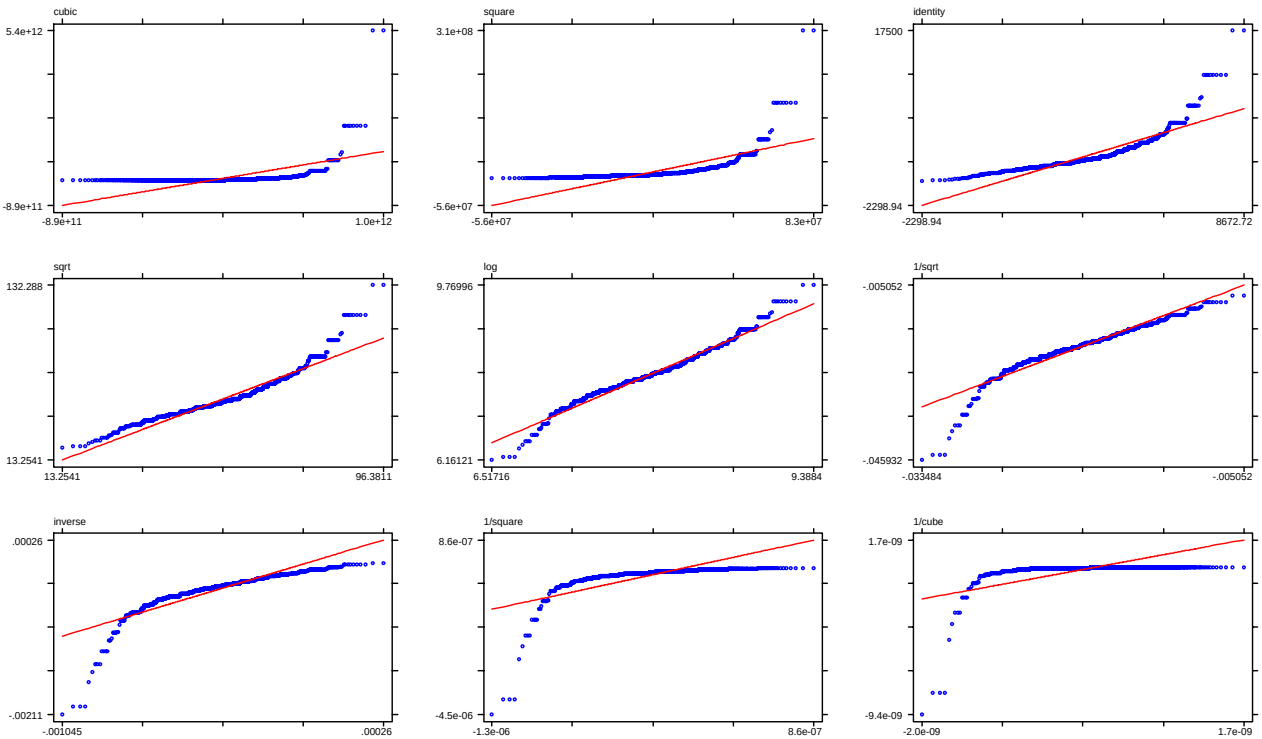
Example: Regression on income (only West-Germans)



Especially at high incomes there is departure from normality (positive skew).

Since we observe heteroscedasticity and nonnormality we should apply a proper transformation. Stata has a nice command that helps here:

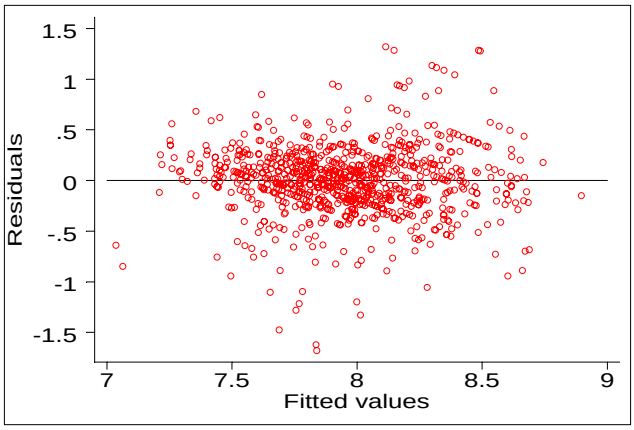
```
qladder income
```

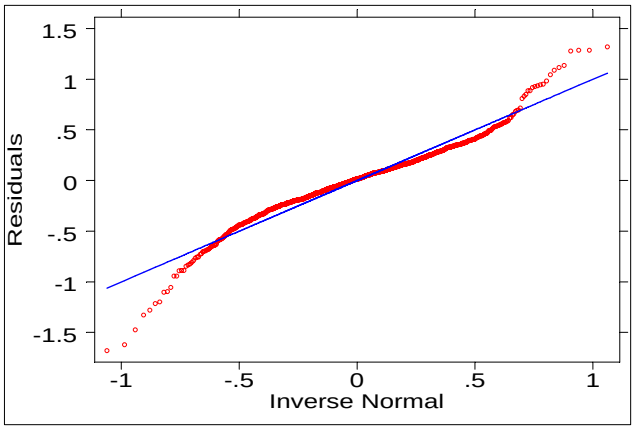


income

Quantile-Normal Plots by Transformation

A log-transformation (q=0) seems best. Using ln(income) as dependent variable we obtain the following plots:





This transformation alleviates our problems. There is no heteroscedasticity and only ”light” nonnormality (heavy tails).

This is our result:

```
. regress lnincome educ exp exp2 prestf woman white civil self

  Source |       SS       df       MS          Number of obs =     849
---------+------------------------------       F(  8,   840) =   82.80
   Model |  81.4123948     8  10.1765493       Prob > F      =  0.0000
Residual |  103.237891   840  .122902251       R-squared     =  0.4409
---------+------------------------------       Adj R-squared =  0.4356
   Total |  184.650286   848  .217747978       Root MSE      =  .35057

 ------------------------------------------------------------------------------
lnincome|      Coef.   Std. Err.       t     P>|t|   95% Conf. Interval]
--------+---------------------------------------------------------------------
educ    |   .0591425   .0054807     10.791   0.000    .048385       .0699
exp     |   .0496282   .0041655     11.914   0.000   .0414522    .0578041
exp2    |  -.0009166   .0000908    -10.092   0.000  -.0010949   -.0007383
prestf  |    .000618   .0004518      1.368   0.172  -.0002689    .0015048
woman   |  -.3577554   .0291036    -12.292   0.000  -.4148798   -.3006311
white   |   .1714642   .0310107      5.529   0.000   .1105966    .2323318
civil   |   .1705233   .0488323      3.492   0.001   .0746757    .2663709
self    |   .2252737   .0442668      5.089   0.000   .1383872    .3121601
_cons   |   6.669825   .0734731     90.779   0.000   6.525613    6.814038
 ------------------------------------------------------------------------------
```

R^2 for the regression on ”income” was 37.7%. Here it is 44.1%. However, it makes no sense to compare both, because the variance to be explained differs between these two variables!

Note that we finally arrived at a specification that is identical to the one derived from human capital theory. Thus, data driven diagnostics support strongly the validity of human capital theory!

Interpretation: The problem with transformations is that interpretation becomes more difficult. In our case we arrived at an semi-logarithmic specification. The standard interpretation of regression coefficients is no longer valid. Now our model is:

$$\ln(y_i) = \beta_0 + \beta_1 x_i + \epsilon_i,$$

or

$$E(y|x) = e^{\beta_0 + \beta_1 x}.$$

Coefficients are effects on ln(income). This nobody can understand. One wants an interpretation in terms of income. The marginal effect on income is

$$\frac{dE(y|x)}{dx} = E(y|x)\beta_1.$$

The discrete (unit) effect on income is

$$E(y|x+1) - E(y|x) = E(y|x)(e^{\beta_1} - 1).$$

Unlike in the linear regression model, both effects are not equal and depend on the value of X! It is generally preferable to use the discrete effect. This, however, can be transformed:

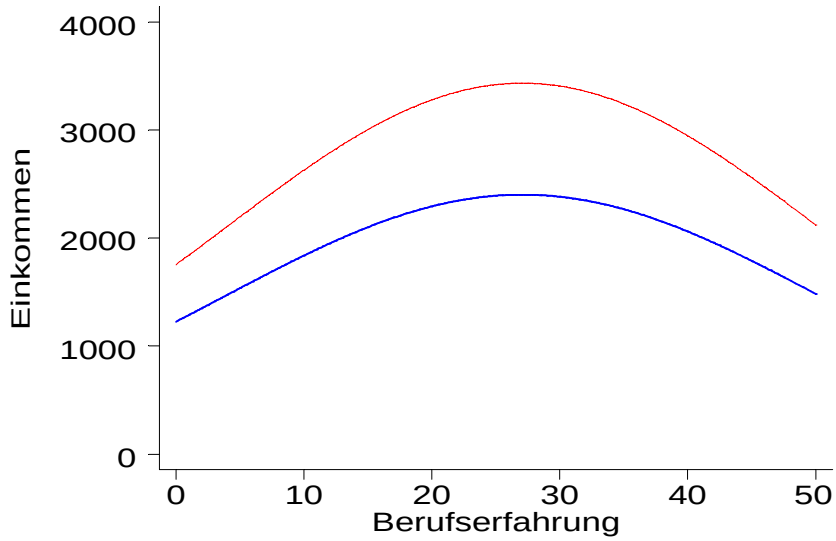
$$\frac{E(y|x+1) - E(y|x)}{E(y|x)} = e^{\beta_1} - 1.$$

This is the percentage change of Y with an unit increase of X. Thus, coefficients of a semi-logarithmic regression can be interpreted as discrete percentage effects (rate of return).

This interpretation is eased further if $\beta_1 < 0.1$, then $e^{\beta_1} - 1 \approx \beta_1$.

Example: For women we have $e^{-.358} - 1 = -.30$. Women's earnings are 30% below men's.

These are percentage effects, don't confuse this with absolute change! Let's produce a conditional-effect plot (prestf=50, educ=13, blue collar).



blue: woman, red: man

Clearly the absolute difference between men and women depends on exp. But the relative difference is constant.

Influential data

A data point is influential if it changes the results of a regression.

Problem: (only in extreme cases). The regression does not ”represent” the majority of cases, but only a few.

Diagnosis: Influence on coefficients=leverage x discrepancy. Leverage is an unusual x-value, discrepancy is ”outlyingness”.

Remedy: Check whether the data point is correct. If yes, then try to improve the specification (are there common characteristics of the influential points?). Don’t throw away influential points (robust regression)! This is data manipulation.

Partial-regression plot

Scattergrams are useful in simple regression. In multiple regression one has to use partial-regression scattergrams (added-variable plot in Stata, avplot). Plot the residual from the regression of Y on all X (without X_j) against the residual from the regression of X_j on the other X. Thus one partials out the effects of the other X-variables.

Influence Statistics

Influence can be measured directly by dropping observations. How changes $\hat{\beta}_j$, if we drop case i ($\hat{\beta}_{j(-i)}$).

$$DFBETAS_{ij} = \frac{\hat{\beta}_j - \hat{\beta}_{j(-i)}}{\hat{\sigma}_{\hat{\beta}_{j(-i)}}}$$

shows the (standardized) influence of case i on coefficient j.

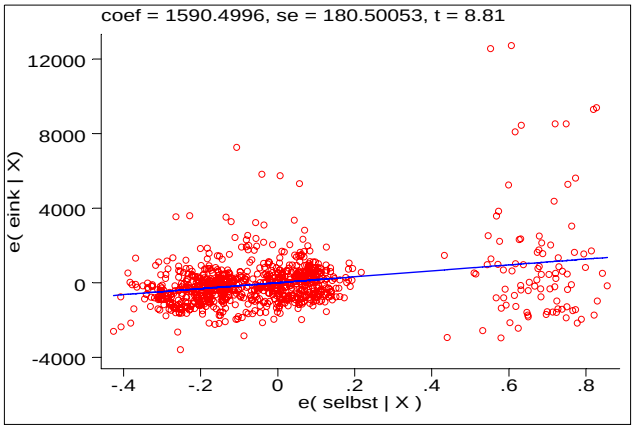
$$\begin{aligned} DFBETAS_{ij} &> 0, \quad \text{case i pulls } \hat{\beta}_j \text{ up} \\ DFBETAS_{ij} &< 0, \quad \text{case i pulls } \hat{\beta}_j \text{ down} \end{aligned} .$$

Influential are cases beyond the cutoff $2/\sqrt{n}$. There is a DFBETAS_{ij} for every case and variable. To judge the cutoff, one should use index-plots.

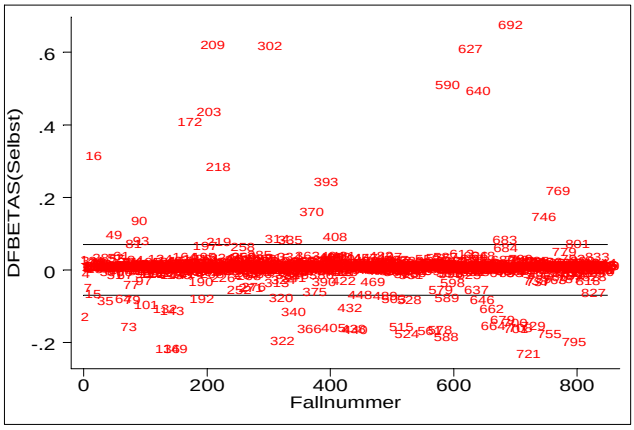
It is easier to use Cook’s D, which is a measure that ”averages” the DFBETAS. The cutoff is here $4/n$.

Example: Regression on income (only West-Germans)

For didactical purposes we use again the regression on income. Let's have a look on the effect of "self".



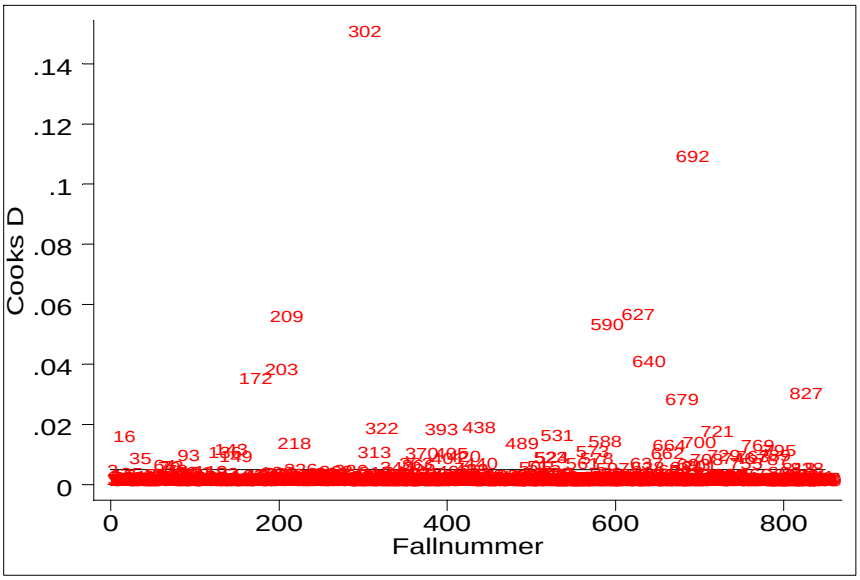
partial-regression plot for "self"



index-plot for DFBETAS(Self)

There are some self-employed persons with high income residuals who pull up the regression line. Obviously the cutoff is much too low.

However, it is easier to have a look on the index-plot for Cook's D.



Again the cutoff is much too low. But we identify two cases, who differ very much from the rest. Let's have a look on these data:

```
         income    yhat         exp         woman      self         D
302.     17500   5808.125       31.5          0          1   .1492927
692.     17500   5735.749       28.5          0          1   .1075122
```

These are two self-employed men, with extremely high income (”above 15.000 DM” is the true value). They exert strong influence on the regression.

What to do? Obviously we have a problem with self-employed people that is not cured by including the dummy. Thus, there is good reason to drop the self-employed from the sample. This is also what theory would tell us. Our final result is then (on ln(income)):

```
  Source |       SS       df       MS          Number of obs =     756
---------+------------------------------       F(  7,   748) =  105.47
   Model |  60.6491102     7  8.66415861       Prob > F      =  0.0000
Residual |  61.4445399   748  .082145107       R-squared     =  0.4967
---------+------------------------------       Adj R-squared =  0.4920
   Total |   122.09365   755  .161713444       Root MSE      =  .28661

 ------------------------------------------------------------------------------
lnincome|      Coef.   Std. Err.       t     P>|t|  [95% Conf. Interval]
 -------+----------------------------------------------------------------
educ    |    .057521   .0047798     12.034   0.000  .0481377    .0669044
exp     |   .0433609   .0037117     11.682   0.000  .0360743    .0506475
exp2    |  -.0007881   .0000834     -9.455   0.000 -.0009517   -.0006245
prestf  |   .0005446   .0003951      1.378   0.168  -.000231    .0013203
woman   |  -.3211721   .0249711    -12.862   0.000  -.370194   -.2721503
white   |   .1630886   .0258418      6.311   0.000  .1123575    .2138197
civil   |   .1790793   .0402933      4.444   0.000  .0999779    .2581807
 _cons  |   6.743215   .0636083    106.012   0.000  6.618343    6.868087
 ------------------------------------------------------------------------
```

Since we changed our specification, we should start anew and test whether regression assumptions also hold for this specification.

4) Binary Response Models

With Y nominal, a mean regression makes no sense. One can, however, investigate conditional relative frequencies. Thus a regression is given by the J+1 functions

$$\pi_j(x) = f(Y = j|X = x) \qquad \text{for } j = 0, 1, \ldots, J.$$

For discrete X this is a cross tabulation! If we have many X and/or continuous X, however, it makes sense to use a parametric model. The function used must have the following properties:

$$\begin{gathered} 0 \leq \pi_0(x;\beta), \ldots, \pi_J(x;\beta) \leq 1 \\ \sum_{j=0}^{J} \pi_j(x;\beta) = 1 \end{gathered} .$$

Therefore, most binary models use distribution functions.

The binary logit model

Y is dichotomous (J=1). We choose the logistic distribution $\Lambda(z) = \exp(z)/(1 + \exp(z))$, so we get the binary logit model (logistic regression). Further, specify a linear model for z ($\beta_0 + \beta_1 x_1 + \ldots + \beta_p x_p = \boldsymbol{\beta}'\mathbf{x}$):

$$P(Y = 1) = \frac{e^{\boldsymbol{\beta}'\mathbf{x}}}{1 + e^{\boldsymbol{\beta}'\mathbf{x}}} = \frac{1}{1 + e^{-\boldsymbol{\beta}'\mathbf{x}}}$$

$$P(Y = 0) = 1 - P(Y = 1) = \frac{1}{1 + e^{\boldsymbol{\beta}'\mathbf{x}}}.$$

Coefficients are not easy to interpret. Below we will discuss this in detail. Here we use only the sign interpretation (positive means P(Y=1) increases with X).

Example 1: party choice and West/East (discrete X)

In the ALLBUS there is as ”Sonntagsfrage” (v329). We dichotomize: CDU/CSU=1, other party=0 (only those, who would vote). We look for the effect of West/East. This is the crosstab:

```
           |          east
       cdu |         0          1 |     Total
-----------+----------------------+----------
         0 |      1043        563 |      1606
           |     66.18      77.98 |     69.89
-----------+----------------------+----------
         1 |       533        159 |       692
           |     33.82      22.02 |     30.11
-----------+----------------------+----------
     Total |      1576        722 |      2298
           |    100.00     100.00 |    100.00
```

This is the result of a logistic regression:

```
. logit cdu east

Iteration 0:   log likelihood = -1405.9621
Iteration 1:   log likelihood = -1389.1023
Iteration 2:   log likelihood = -1389.0067
Iteration 3:   log likelihood = -1389.0067

Logit estimates                                   Number of obs   =       2298
                                                  LR chi2(1)      =      33.91
                                                  Prob > chi2     =     0.0000
Log likelihood = -1389.0067                       Pseudo R2       =     0.0121

------------------------------------------------------------------------------
 cdu |      Coef.   Std. Err.     z     P>|z|   [95% Conf. Interval]
-----+------------------------------------------------------------------------
east |  -.5930404   .1044052   -5.680   0.000  -.7976709   -.3884099
cons |   -.671335   .0532442  -12.609   0.000  -.7756918   -.5669783
------------------------------------------------------------------------------
```

The negative coefficient tells us, that East-Germans vote less often for CDU (significantly). However, this only reproduces the crosstab in a complicated way:

$$P(Y = 1|X = East) = \frac{1}{1 + e^{-(-.671-.593)}} = .220$$

$$P(Y = 1|X = West) = \frac{1}{1 + e^{-(-.671)}} = .338.$$

Thus, the logistic brings an advantage only in multivariate models.

Why not OLS?

It is possible to estimate an OLS regression with such data:

$$E(Y|\mathbf{x}) = P(Y = 1|\mathbf{x}) = \boldsymbol{\beta}'\mathbf{x}.$$

This is the linear probability model. It has, however, nonnormal and heteroscedastic residuals. Further, prognoses can be beyond $[0, 1]$. Nevertheless, it often works pretty well.

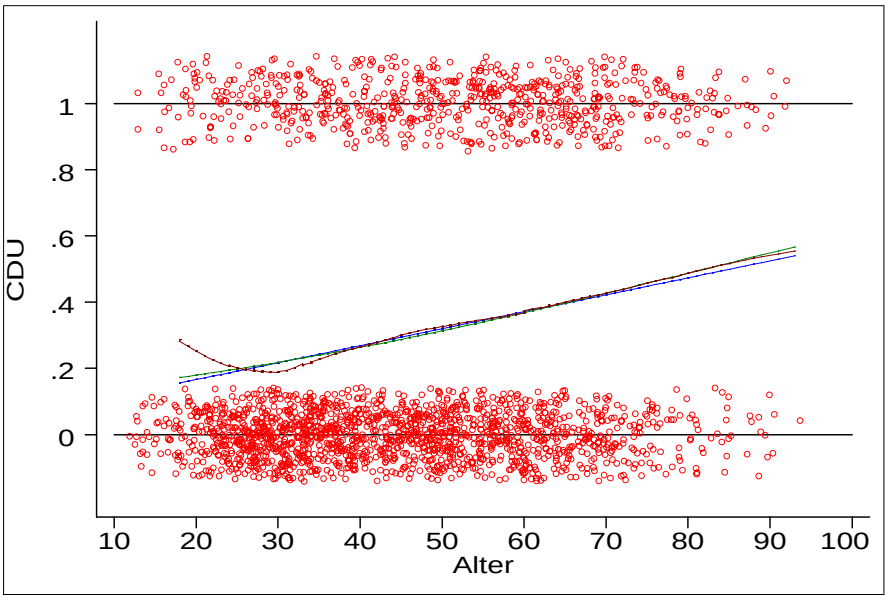
```
. regr cdu east
R-squared     =  0.0143

------------------------------------------------------------------------
 cdu |      Coef.   Std. Err.       t     P>|t|    [95% Conf. Interval]
-----+------------------------------------------------------------------
east |  -.1179764   .0204775     -5.761   0.000   -.1581326   -.0778201
cons |    .338198   .0114781     29.465   0.000    .3156894    .3607065
------------------------------------------------------------------------
```

It gives a discrete effect on P(Y=1). This is exactly the percentage point difference from the crosstab. Given the ease of interpretation of this model, one should not discard it from the beginning.

Example 2: party choice and age (continuous X)

```
. logit   cdu age

Iteration 0:   log likelihood = -1405.2452
Iteration 3:   log likelihood = -1364.6916

Logit estimates                   Number of obs   =       2296
                                  LR chi2(1)      =      81.11
                                  Prob > chi2     =     0.0000
Log likelihood = -1364.6916       Pseudo R2       =     0.0289

-------------------------------------------------------
     cdu |      Coef.   Std. Err.       z     P>|z|
---------+---------------------------------------------
   age   |   .0245216    .002765      8.869   0.000
   _cons |  -2.010266   .1430309    -14.055   0.000
-------------------------------------------------------

. regress cdu age
R-squared     =  0.0353
-------------------------------------------------------
     cdu |      Coef.   Std. Err.       t     P>|t|
---------+---------------------------------------------
   age   |   .0051239    .000559      9.166   0.000
   _cons |   .0637782   .0275796      2.313   0.021
-------------------------------------------------------
```

With age P(CDU) increases. The linear model says the same.



This is a (jittered) scattergram of the data with estimated regression lines: OLS (blue), logit (green), lowess (brown). They are almost identical. The reason is that the logistic function is almost linear in interval $[0.2, 0.8]$. Lowess hints towards a nonmonotone effect at young ages (this is a diagnostic plot to detect deviations from the logistic function).

Interpretation of logit coefficients

There are many ways to interpret the coefficients of a logistic regression. This is due to the nonlinear nature of the model.

Effects on a latent variable

It is possible to formulate the logit model as a threshold model with a continuous, latent variable Y^*. Example from above: Y^* is the (unobservable) utility difference between CDU and other parties. We specify a linear regression model for Y^*:

$$y^* = \boldsymbol{\beta}'\mathbf{x} + \epsilon,$$

We do not observe Y^*,but only the resulting binary choice variable Y that results form the following threshold model:

$$y = 1, \quad \text{for} \quad y^* > 0,$$

$$y = 0, \quad \text{for} \quad y^* \leq 0.$$

To make the model practical, one has to assume a distribution for ϵ. With the logistic distribution, we obtain the logit model.

Thus, logit coefficients could be interpreted as discrete effects on Y^*. Since the scale of Y^* is arbitrary, this interpretation is not useful.

Note: It is erroneous to state that the logit model contains no error term. This becomes obvious if we formulate the logit as threshold model on a latent variable.

Probabilities, odds, and logits

Let's now assume a continuous X. The logit model has three equivalent forms:

Probabilities:

$$P(Y = 1|x) = \frac{e^{\alpha+\beta x}}{1 + e^{\alpha+\beta x}}.$$

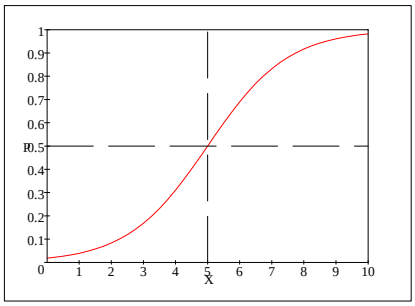
Odds:

$$\frac{P(Y = 1|x)}{P(Y = 0|x)} = e^{\alpha+\beta x}.$$

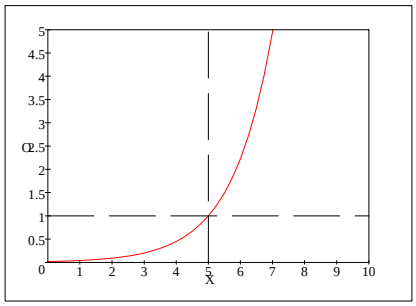
Logits (Log-Odds):

$$\ln\left(\frac{P(Y = 1|x)}{P(Y = 0|x)}\right) = \alpha + \beta x.$$

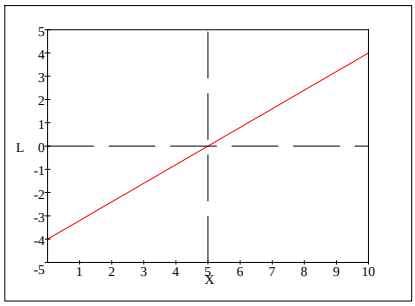
Example: For these plots $\alpha = -4, \quad \beta = 0.8$:



probability



odd



logit

Logit interpretation

β is the discrete effect on the logit. Most people, however, do not understand what a change in the logit means.

Odds interpretation

e^{β} is the (multiplicative) discrete effect on the odds ($e^{\alpha+\beta(x+1)} = e^{\alpha+\beta x}e^{\beta}$). Odds are also not easy to understand, nevertheless this is the standard interpretation in the literature.

Example 1: $e^{-.593} = .55$. The Odds CDU vs. Others is in the East smaller by the factor 0.55:

$Odds_{east} = .22/.78 = .282$,

$Odds_{west} = .338/.662 = .510$,

thus $.510 \cdot .55 = .281$.

Note: Odds are difficult to understand. This leads to often erroneous interpretations. in the example the odds are smaller by about half, not P(CDU)!

Example 2: $e^{.0245} = 1.0248$. For every year the odds increase by 2.5%. In 10 years they increase by 25%? No, because $e^{.0245 \cdot 10} = 1.0248^{10} = 1.278$.

Probability interpretation

This is the most natural interpretation, since most people have an intuitive understanding of what a probability is. The drawback is, however, that these effects depend on the X-value (see plot above). Therefore, one has to choose a value (usually $\overline{x}$) at which to compute the discrete probability effect

$$P(Y = 1|\overline{x} + 1) - P(Y = 1|\overline{x}) = \frac{e^{\alpha+\beta(\overline{x}+1)}}{1 + e^{\alpha+\beta(\overline{x}+1)}} - \frac{e^{\alpha+\beta\overline{x}}}{1 + e^{\alpha+\beta\overline{x}}}.$$

Normally you would have to calculate this by hand, however Stata has a nice ado.

Example 1: The discrete effect is $.338 - .220 = -.118$, i.e. -12 percentage points.

Example 2: Mean age is 46.374. Therefore

$$\frac{1}{1 + e^{2.01-.0245 \cdot 47.374}} - \frac{1}{1 + e^{2.01-.0245 \cdot 46.374}} = 0.00512.$$

The 47. year increases P(CDU) by 0.5 percentage points.

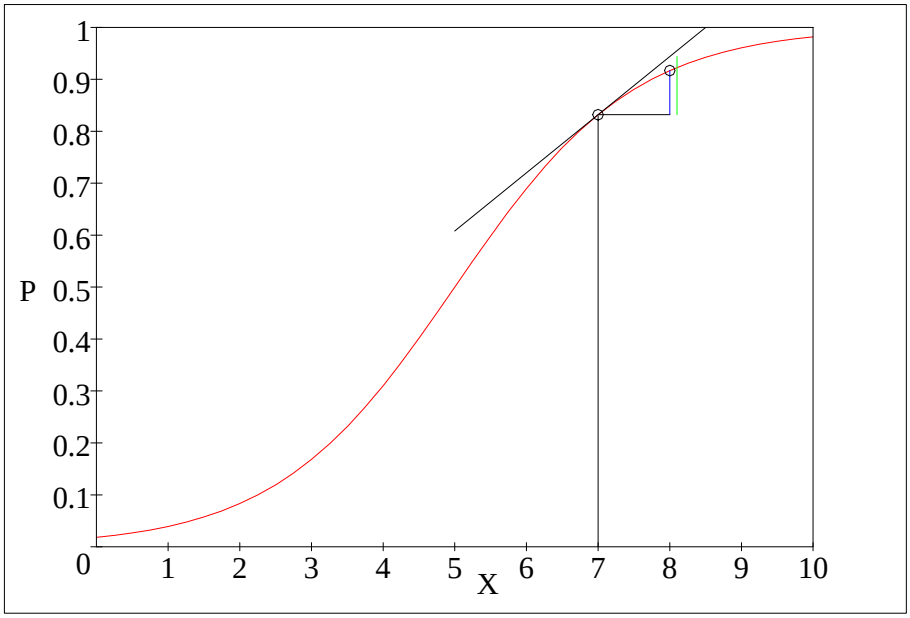
Note: The linear probability model coefficients are identical with these effects!

Marginal effects

Stata computes marginal probability effects. These are easier to compute, but they are only approximations to the discrete effects. For the logit model

$$\frac{\partial P(Y=1|\overline{x})}{\partial x} = \frac{e^{\alpha+\beta\overline{x}}}{(1+e^{\alpha+\beta\overline{x}})^2}\beta = \widehat{P}(Y=1|\overline{x})\widehat{P}(Y=0|\overline{x})\beta.$$

Example: $\alpha = -4, \quad \beta = 0,8, \ \overline{x} = 7$



$\widehat{P}(Y=1|7) = \frac{1}{1+e^{-(-4+0.8\cdot 7)}} =.832, \quad \widehat{P}(Y=1|8) = \frac{1}{1+e^{-(-4+0.8\cdot 8)}} = 0.917$

discrete: $0.917 - 0.832 =.085$

marginal: $0.832 \cdot (1-0.832) \cdot 0.8 =.112$

ML estimation

We have data (y_i, x_i) and a regression model $f(Y=y|X=x;\ \boldsymbol{\beta})$. We want to estimate the parameter $\boldsymbol{\beta}$ in such a way that the model fits the data "best". There are different criteria to do this. The best known is maximum likelihood (ML).

The idea is to choose the $\widehat{\boldsymbol{\beta}}$ that maximizes the likelihood of the data. Given the model and independent draws from it the likelihood is:

$$L(\boldsymbol{\beta}) = \prod_{i=1}^{n} f(y_i, x_i;\ \boldsymbol{\beta}).$$

The ML estimate results from maximizing this function. For computational reasons it is better to maximize the log likelihood:

$$l(\boldsymbol{\beta}) = \sum_{i=1}^{n} \ln f(y_i, x_i;\ \boldsymbol{\beta}).$$

Compute the first derivatives and set them equal 0.

ML estimates have some desirable statistical properties (asymptotic).

- consistent: $\mathrm{E}(\widehat{\boldsymbol{\beta}}_{\mathrm{ML}}) = \boldsymbol{\beta}$
- normally distributed: $\widehat{\boldsymbol{\beta}}_{\mathrm{ML}} \sim \mathrm{N}(\boldsymbol{\beta}, \mathbf{I}(\boldsymbol{\beta})^{-1})$, where $\mathbf{I}(\boldsymbol{\beta}) = -\mathrm{E}(\frac{\partial^2 \ln \mathrm{L}}{\partial \boldsymbol{\beta} \partial \boldsymbol{\beta}'})$
- efficient: ML estimates obtain minimal variance (Rao-Cramer)

ML estimates for the binary logit model

The probability to observe a data point with Y=1 is P(Y=1). Accordingly for Y=0. Thus the likelihood is

$$L(\boldsymbol{\beta}) = \prod_{i=1}^{n} \left\{ \left[\frac{e^{\boldsymbol{\beta}'\mathbf{x}_i}}{1+e^{\boldsymbol{\beta}'\mathbf{x}_i}} \right]^{y_i} \cdot \left[\frac{1}{1+e^{\boldsymbol{\beta}'\mathbf{x}_i}} \right]^{(1-y_i)} \right\}.$$

The log likelihood is

$$\begin{aligned} l(\boldsymbol{\beta}) &= \sum_{i=1}^{n} \left\{ y_i \ln\left[\frac{e^{\boldsymbol{\beta}'\mathbf{x}_i}}{1+e^{\boldsymbol{\beta}'\mathbf{x}_i}} \right] + (1-y_i) \ln\left[\frac{1}{1+e^{\boldsymbol{\beta}'\mathbf{x}_i}} \right] \right\} \\ &= \sum_{i=1}^{n} y_i \boldsymbol{\beta}'\mathbf{x}_i - \sum_{i=1}^{n} \ln(1+e^{\boldsymbol{\beta}'\mathbf{x}_i}). \end{aligned}$$

Taking derivatives yields:

$$\frac{\partial l(\boldsymbol{\beta})}{\partial \boldsymbol{\beta}} = \sum y_i \mathbf{x}_i - \sum \frac{e^{\boldsymbol{\beta}'\mathbf{x}_i}}{1+e^{\boldsymbol{\beta}'\mathbf{x}_i}} \mathbf{x}_i.$$

Setting equal to 0 yields the estimation equations:

$$\sum y_i \mathbf{x}_i = \sum \frac{e^{\widehat{\boldsymbol{\beta}}'\mathbf{x}_i}}{1+e^{\widehat{\boldsymbol{\beta}}'\mathbf{x}_i}} \mathbf{x}_i.$$

These equations have no closed form solution. One has to solve them by iterative numerical algorithms.

Significance tests and model fit

Overall significance test

Compare the log likelihood of the full model ($\ln L_1$) with the one from the constant only model ($\ln L_0$). Compute the likelihood ratio test statistic:

$$\chi^2 = -2\ln\left(\frac{L_0}{L_1}\right) = 2(\ln L_1 - \ln L_0).$$

Under the null $H_0 : \beta_1 = \beta_2 = \ldots = \beta_p = 0$ this statistic is distributed asymptotically χ^2_p.

Example 2: $\ln L_1 = -1364.7$ and $\ln L_0 = -1405.2$ (Iteration 0).

$$\chi^2 = 2(-1364.7 + 1405.2) = 81.0.$$

With one degree of freedom we can reject the H_0.

Testing one coefficient

Compute the z-value (coefficient/S.E.) which is distributed asymptotically normally.

One could also use the LR-test (this test is ”better”). Use also the LR-test to test restrictions on a set of coefficients.

Model fit

With nonmetric Y we no longer can define a unique measure of fit like R^2(this is due to the different conceptions of variation in nonmetric models). Instead there are many pseudo-R^2 measures. The most popular one is *McFadden’s Pseudo-*R^2:

$$R^2_{MF} = \frac{\ln L_0 - \ln L_1}{\ln L_0}.$$

Experience tells that it is ”conservative”. Another one is *McKelvey-Zavoina’s Pseudo-*R^2 (formula see Long, p. 105). This measure is suggested by the authors of several simulation studies, because it most closely approximates the R^2 obtained from regressions on the underlying latent variable.

A completely different approach has been suggested by Raftery (see Long, pp. 110). He favors the use of the Bayesian information criterion (BIC). This measure can also be used to compare non-nested models!

An example using Stata

We continue our party choice model by adding education, occupation, and sex (output changed by inserting odds ratios and marginal effects).

```
. logit    cdu educ age east woman white civil self trainee

Iteration 0:   log likelihood = -757.23006
Iteration 1:   log likelihood = -718.71868
Iteration 2:   log likelihood = -718.25208
Iteration 3:   log likelihood = -718.25194

Logit estimates                   Number of obs   =       1262
                                  LR chi2(8)      =      77.96
                                  Prob > chi2     =     0.0000
Log likelihood = -718.25194       Pseudo R2       =     0.0515
------------------------------------------------------------------
   cdu |     Coef.   Std. Err.    z     P>|z|  Odds Ratio  MargEff
-------+----------------------------------------------------------
educ   |   -.04362   .0264973  -1.646   0.100    .9573177  -0.0087
age    |  .0351726   .0059116   5.950   0.000   1.035799    0.0070
east   | -.4910153   .1510739  -3.250   0.001    .6120047  -0.0980
woman  | -.1647772   .1421791  -1.159   0.246    .8480827  -0.0329
white  |  .1342369   .1687518   0.795   0.426   1.143664    0.0268
civil  |   .396132   .2790057   1.420   0.156   1.486066    0.0791
self   |  .6567997   .2148196   3.057   0.002   1.92861     0.1311
trainee|  .4691257   .4937517   0.950   0.342   1.598596    0.0937
 _cons | -1.783349   .4114883  -4.334   0.000
------------------------------------------------------------------
```

Thanks to Scott Long there are several helpful ados:

```
. fitstat

Measures of Fit for logit of cdu

Log-Lik Intercept Only:     -757.230   Log-Lik Full Model:   -718.252
D(1253):                    1436.504   LR(8):                  77.956
                                       Prob > LR:               0.000
McFadden's R2:                 0.051   McFadden's Adj R2:       0.040
Maximum Likelihood R2:         0.060   Cragg & Uhler's R2:      0.086
McKelvey and Zavoina's R2:     0.086   Efron's R2:              0.066
Variance of y*:                3.600   Variance of error:       3.290
Count R2:                      0.723   Adj Count R2:            0.039
AIC:                           1.153   AIC*n:                1454.504
BIC:                        -7510.484  BIC':                  -20.833


. prchange, help

logit: Changes in Predicted Probabilities for cdu

        min->max      0->1     -+1/2    -+sd/2  MargEfct
educ     -0.1292   -0.0104   -0.0087   -0.0240   -0.0087
aage      0.4271    0.0028    0.0070    0.0808    0.0070
east     -0.0935   -0.0935   -0.0978   -0.0448   -0.0980
```

```
woman     -0.0326   -0.0326   -0.0329   -0.0160   -0.0329
white      0.0268    0.0268    0.0268    0.0134    0.0268
civil      0.0847    0.0847    0.0790    0.0198    0.0791
self       0.1439    0.1439    0.1307    0.0429    0.1311
trainee    0.1022    0.1022    0.0935    0.0138    0.0937
```

Diagnostics

Perfect discrimination

If a X perfectly discriminates between Y=0 and Y=1 the logit will be infinite and the resp. coefficient goes towards infinity. Stata drops this variable automatically (other programs do not!).

Functional form

Use scattergram with lowess (see above).

Influential data

We investigate not single cases but X-patterns. There are K patterns, m_k is the number of cases with pattern k. $\widehat{P}_k$ is the predicted $P(Y = 1)$ and Y_k is the number of ones.

Pearson residuals are defined by

$$r_k = \frac{Y_k - m_k\widehat{P}_k}{\sqrt{m_k\widehat{P}_k(1 - \widehat{P}_k)}}.$$

The Pearson χ^2 statistic is

$$\chi^2 = \sum_{k=1}^{K} r_k^2.$$

This measures the deviation from the saturated model (this is a model that contains a parameter for every X-pattern). The saturated model fits the data perfectly (see example 1).

Using Pearson residuals we can construct measures of influence. $\Delta\chi^2_{(-k)}$ measures the decrease in χ^2, if we drop pattern k

$$\Delta\chi^2_{(-k)} = \frac{r_k^2}{1 - h_k}.$$

$h_k = m_k h_i,$ where h_i is an element from the hat matrix. Large

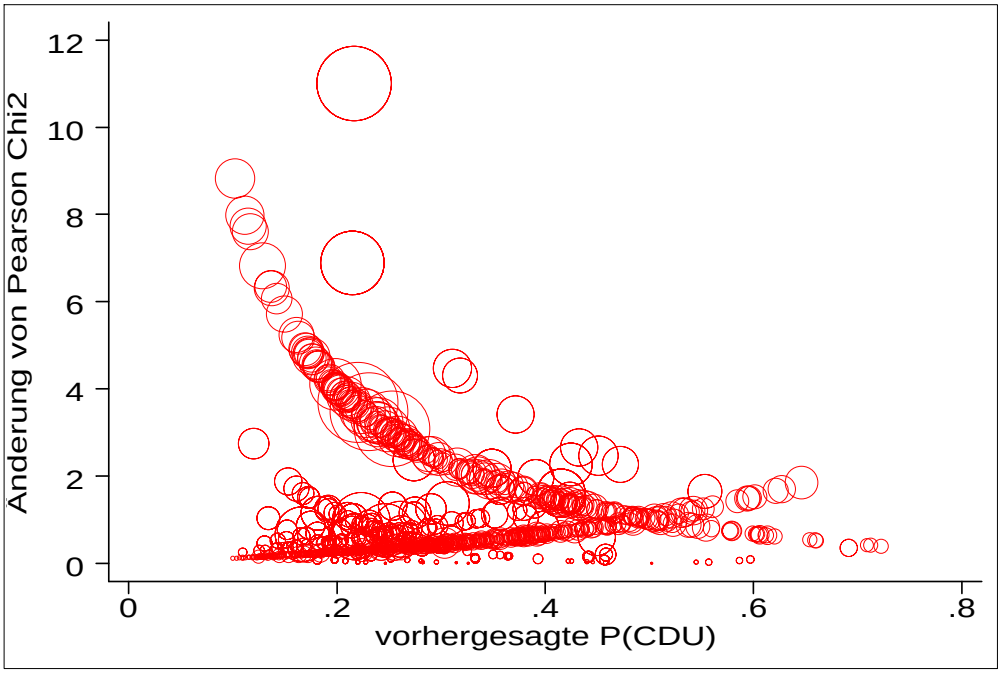
values of $\Delta\chi^2_{(-k)}$ indicate that the model would fit much better, if pattern k would be dropped.

A second measure is constructed in analogy with Cook's D and measures the standardized change of the logit coefficients, if pattern k would be dropped:

$$\Delta B_{(-k)} = \frac{r_k^2 h_k}{(1-h_k)^2}.$$

A large value of $\Delta B_{(-k)}$ shows that pattern k exerts influence on the estimation results.

Example: We plot $\Delta\chi^2_{(-k)}$ against $\widehat{P}_k$. Circles proportional to $\Delta B_{(-k)}$.



One should spend some thoughts on the patterns that have large circles and are high up. If one lists these patterns one can see that these are young woman who vote for CDU. The reason might be the nonlinearity at young ages that we observed earlier. We could model this by adding a "young voters" dummy.

The binary probit model

We obtain the probit model, if we specify a normal error distribution for the latent variable model. The resulting probability model is

$$P(Y = 1) = \Phi(\beta'\mathbf{x}) = \int_{-\infty}^{\beta'\mathbf{x}} \phi(t)\, dt.$$

The practical disadvantage is that it is hard to calculate probabilities by hand. We can apply all procedures from above analogously (only the odds interpretation does not work). Since logistic and normal distribution are very similar, results are in most situations identical for all practical purposes. Coefficients can be transformed by a scaling factor (multiply probit coefficients by 1.6-1.8). Only in the tails results may be different.

5) The Multinomial Logit Model

$J > 1$ and using the multivariate logistic distribution we get

$$\pi_j(\boldsymbol{\beta}_j'\mathbf{x}) = \frac{\exp(\boldsymbol{\beta}_j'\mathbf{x})}{\sum_{k=0}^{J}\exp(\boldsymbol{\beta}_k'\mathbf{x})}.$$

One of these functions is redundant since they must sum to 1. We normalize with $\boldsymbol{\beta}_0 = \mathbf{0}$ and obtain the multinomial logit model

$$P(Y = j|\mathbf{X} = \mathbf{x}) = \frac{e^{\boldsymbol{\beta}_j'\mathbf{x}}}{1 + \sum_{k=1}^{J} e^{\boldsymbol{\beta}_k'\mathbf{x}}}, \quad \text{for } j = 1, 2, \ldots, J$$

$$P(Y = 0|\mathbf{X} = \mathbf{x}) = \frac{1}{1 + \sum_{k=1}^{J} e^{\boldsymbol{\beta}_k'\mathbf{x}}}.$$

The binary logit model is a special case for $J = 1$. Estimation is done by ML.

Example 1: Party choice and West/East (discrete X)

We distinguish 6 parties: others=0, CDU=1, SPD=2, FDP=3, Grüne=4, PDS=5.

party	east 0	east 1	Total
others	82	31	113
	5.21	4.31	4.93
CDU	533	159	692
	33.88	22.11	30.19
SPD	595	258	853
	37.83	35.88	37.22
FDP	135	65	200
	8.58	9.04	8.73
Gruene	224	91	315
	14.24	12.66	13.74
PDS	4	115	119
	0.25	15.99	5.19
Total	1573	719	2292
	100.00	100.00	100.00

```
. mlogit   party east, base(0)

Iteration 0:   log likelihood =  -3476.897
....
Iteration 6:   log likelihood = -3346.3997

Multinomial regression         Number of obs   =       2292
                               LR chi2(5)      =     260.99
                               Prob > chi2     =     0.0000
Log likelihood = -3346.3997    Pseudo R2       =     0.0375

------------------------------------------------------
  party |      Coef.   Std. Err.       z     P>|z|
---------+--------------------------------------------
CDU      |
    east |  -.2368852   .2293876     -1.033   0.302
   _cons |   1.871802   .1186225     15.779   0.000
---------+--------------------------------------------
SPD      |
    east |   .1371302   .2236288      0.613   0.540
   _cons |   1.981842   .1177956     16.824   0.000
---------+--------------------------------------------
FDP      |
    east |   .2418445   .2593168      0.933   0.351
   _cons |   .4985555    .140009      3.561   0.000
---------+--------------------------------------------
Gruene   |
    east |   .0719455    .244758      0.294   0.769
   _cons |   1.004927   .1290713      7.786   0.000
---------+--------------------------------------------
PDS      |
    east |    4.33137   .5505871      7.867   0.000
   _cons |  -3.020425   .5120473     -5.899   0.000
------------------------------------------------------
(Outcome party==others is the comparison group)
```

Comparing with the crosstab we see that the sign interpretation is no longer correct! For instance would we infer that East-Germans have a higher probability of voting SPD. This, however, is not true as can be seen from the crosstab.

Interpretation of multinomial logit coefficients

Logit interpretation

We denote $P(Y = j)$ by P_j, then

$$\ln\left(\frac{P_j}{P_0}\right) = \boldsymbol{\beta}_j^{'}\mathbf{x}.$$

This is similar to the binary model and not very helpful.

Odds interpretation

The multinomial formulated in terms of the odds is

$$\frac{P_j}{P_0} = e^{\boldsymbol{\beta}_j'\mathbf{x}}.$$

$e^{\beta_{jk}}$ is the (multiplicative) discrete effect of variable X_k on the odds. The sign of β_{jk} gives the sign of the odds effect. They are not easy to understand, but they do not depend on the values of **X**.

Example 1: The odds effect for SPD is $e^{.137} = 1.147$.

Odds$_{east}$ $=.359/.043 = 8.35$,

Odds$_{west}$ $=.378/.052 = 7.27$,

thus $8.35/7.27 = 1.149$.

Probability interpretation

There is a formula to compute marginal effects

$$\frac{\partial P_j}{\partial \mathbf{x}} = P_j\left(\boldsymbol{\beta}_j - \sum_{k=1}^{J} P_k \boldsymbol{\beta}_k\right).$$

The marginal effect clearly depends on **X**. It is common to evaluate this formula at the mean of **X** (possibly dummies set to 0 or 1). Further, it becomes clear that the sign of the marginal effect can be different from the sign of the logit coefficient. It might even be the case that the marginal effect changes sign while **X** changes! Clearly, we should compute them at different **X**-values, or even better, produce conditional effect plots.

Stata computes marginal effects. But they approximate the discrete effects only, and if some $P(Y = j|\mathbf{x})$ are below 0.1 or above 0.9 the approximation is bad. Stata has also an ado by Scott Long that computes discrete effects. Thus, it is better to compute these. However, keep in mind that the discrete effects also depend on the **X**-value.

Example: A multivariate multinomial logit model

We include as independent variables age, education, and West/East (constants are dropped from the output).

```
. mlogit    party educ age east, base(0)

Iteration 0:   log likelihood =  -3476.897

Iteration 6:   log likelihood = -3224.9672

Multinomial regression                  Number of obs   =       2292
                                        LR chi2(15)     =     503.86
                                        Prob > chi2     =     0.0000
Log likelihood = -3224.9672             Pseudo R2       =     0.0725

------------------------------------------------------------
       party |      Coef.   Std. Err.      z    P>|z|
-------------+----------------------------------------------
CDU          |
        educ |    .157302   .0496189     3.17   0.002
         age |   .0437526   .0065036     6.73   0.000
        east |  -.3697796   .2332663    -1.59   0.113
-------------+----------------------------------------------
SPD          |
        educ |   .1460051   .0489286     2.98   0.003
         age |   .0278169    .006379     4.36   0.000
        east |   .0398341   .2259598     0.18   0.860
-------------+----------------------------------------------
FDP          |
        educ |   .2160018   .0535364     4.03   0.000
         age |   .0215305   .0074899     2.87   0.004
        east |   .1414316   .2618052     0.54   0.589
-------------+----------------------------------------------
Gruene       |
        educ |   .2911253   .0508252     5.73   0.000
         age |  -.0106864   .0073624    -1.45   0.147
        east |   .0354226   .2483589     0.14   0.887
-------------+----------------------------------------------
PDS          |
        educ |   .2715325   .0572754     4.74   0.000
         age |   .0240124    .008752     2.74   0.006
        east |   4.209456   .5520359     7.63   0.000
------------------------------------------------------------
(Outcome party==other is the comparison group)
```

There are some quite strong effects (judged by the z-value). All educ odds-effects are positive. This means that the odds of all parties compared with other increase with education. It is, however, wrong to infer from this that the resp. probabilities increase! For some of these parties the probability effect of education is negative (see below). The odds increase nevertheless, because the probability of voting for other

decreases even stronger with education (the rep-effect!).

First, we compute marginal effects at the mean of the variables (only for SPD shown, add ”nose” to reduce computation time).

```
. mfx compute, predict(outcome(2))

Marginal effects after mlogit
      y  = Pr(party==2) (predict, outcome(2))
         =  .41199209
------------------------------------------------
variable |      dy/dx    Std. Err.     z    P>|z|
---------+--------------------------------------
    educ |  -.0091708       .0042   -2.18   0.029
     age |   .0006398      .00064    1.00   0.319
    east*|  -.0216788      .02233   -0.97   0.332
------------------------------------------------
(*) dy/dx is for discrete change of dummy variable from 0 to 1
```

Note that P(SPD)=0.41. Thus, marginal effects should be good approximations. The effect of educ is negative, contrary to the positive odds effect!

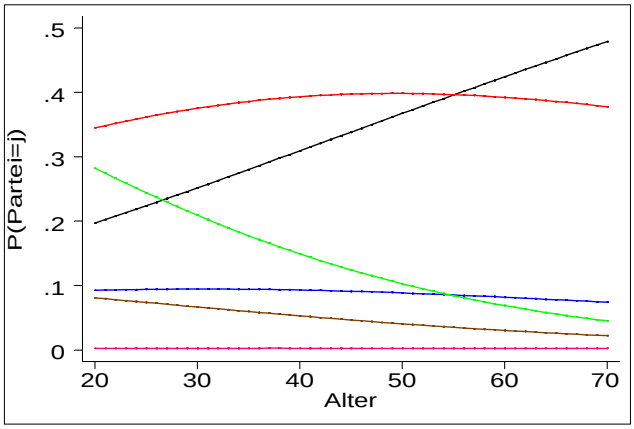
Next, we compute the discrete effects (only for educ shown):

```
. prchange, help

mlogit: Changes in Predicted Probabilities for party

educ
            Avg|Chg|          CDU          SPD          FDP       Gruene
Min->Max   .13715207  -.11109132  -.20352574   .05552502   .33558132
   -+1/2   .00680951  -.00345218  -.00916708    .0045845   .01481096
  -+sd/2   .01834329  -.00927532  -.02462697   .01231783   .03993018
MargEfct   .04085587   -.0034535   -.0091708   .00458626    .0148086

                 PDS       other
Min->Max   .02034985  -.09683915
   -+1/2   .00103305  -.00780927
  -+sd/2   .00278186  -.02112759
MargEfct   .00103308  -.00780364
```

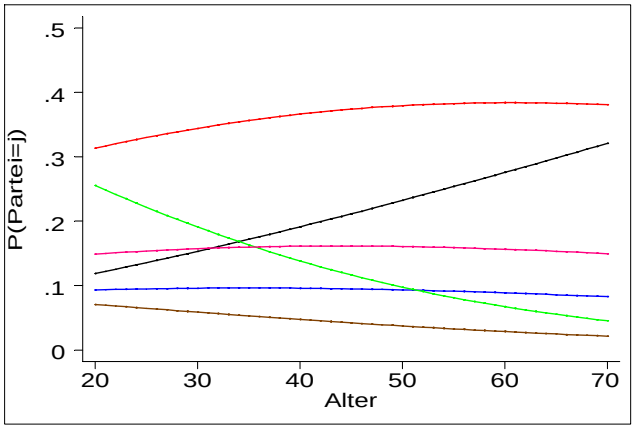
These effects are computed at the mean of **X**. Note that the discrete (and also marginal) effects sum to zero.

To get a complete overview of what is going on in the model, we use conditional effect plots.

First by age (education=12):

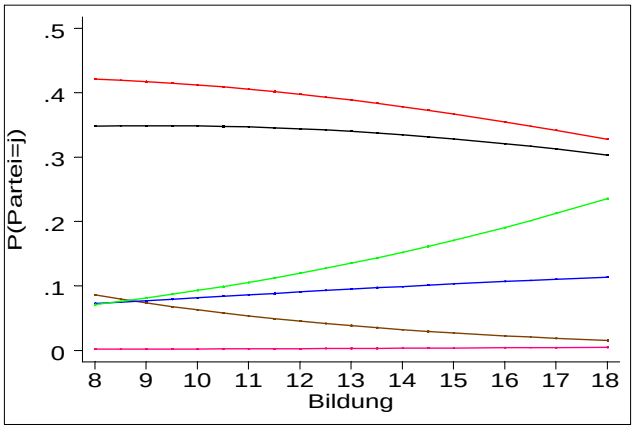


West

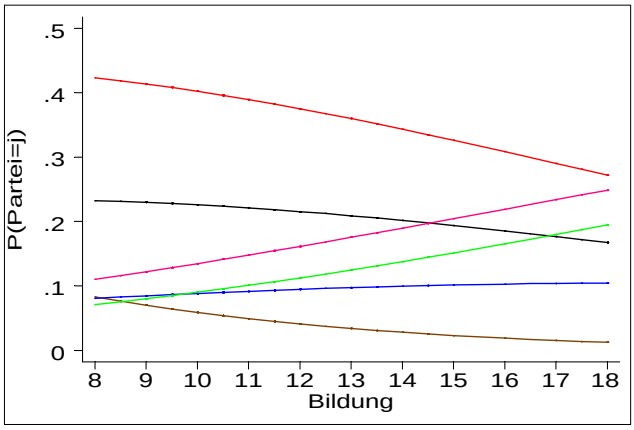


East

Then by education (age=46):



West



East

Other (brown), CDU (black), SPD (red), FDP (blue), Grüne (green), PDS (violet).

Here we see many things. For instance, education effects are positive for three parties (Grüne, FDP, PDS), and negative for the rest. Especially strong is the negative effect on other. This produces the positive odds effects.

Note that the age effect on SPD in the West is non monotonic!

Note: We specified a model without interactions. This is true for the logit effects. But the probability effects show interactions: Look at the effect of education in West and East on the probability for PDS! This is a general point for logit models: though you specify no interactions for logits there might be some in probabilities. The same is also true vice versa. therefore, the only way to make sense out of (multinomial) results are conditional effect plots.

Here are the Stata commands:

```
prgen age, from(20) to(70) x(east 0) rest(grmean) gen(w)
gr7  wp1 wp2 wp3 wp4 wp5 wp6 wx, c(llllll) s(iiiiii) ylabel(0(.1).5)
         xlabel(20(10)70) l1(”P(party=j)”) b2(age) gap(3)
```

Significance tests and model fit

The fit measures work the same way as in the binary model. Not all of them are available.

```
. fitstat

Measures of Fit for mlogit of party

Log-Lik Intercept Only:    -3476.897 Log-Lik Full Model:        -3224.967
D(2272):                    6449.934 LR(15):                      503.860
                                     Prob > LR:                     0.000
McFadden’s R2:                 0.072 McFadden’s Adj R2:             0.067
Maximum Likelihood R2:         0.197 Cragg & Uhler’s R2:            0.207
Count R2:                      0.396 Adj Count R2:                  0.038
AIC:                           2.832 AIC*n:                      6489.934
BIC:                       -11128.939 BIC’:                      -387.802
```

For testing whether a variable is significant we need a LR-Test:

```
. mlogtest, lr

**** Likelihood-ratio tests for independent variables

 Ho: All coefficients associated with given variable(s) are 0.

   party |      chi2    df   P>chi2
---------+-------------------------
    educ |     66.415    5    0.000
     age |    164.806    5    0.000
    east |    255.860    5    0.000
-----------------------------------
```

Though some logit effects were not significant, all three variables show an overall significant effect.

Finally, we can use BIC to compare non nested models. The model with the lower BIC is preferable. An absolute BIC difference of greater 10 is very strong evidence for this model.

```
mlogit    party educ age woman, base(0)
fitstat,  saving(mod1)
mlogit    party educ age east, base(0)
fitstat,  using(mod1)

Measures of Fit for mlogit of party

                              Current            Saved       Difference
Model:                         mlogit           mlogit
N:                               2292             2292                0
Log-Lik Intercept Only:    -3476.897        -3476.897            0.000
Log-Lik Full Model:        -3224.967        -3344.368          119.401
```

```
LR:                          503.860(15)      265.057(15)   238.802(0)
McFadden's R2:                 0.072            0.038           0.034
Adj Count R2:                  0.038            0.021           0.017
BIC:                       -11128.939       -10890.136        -238.802
BIC':                        -387.802         -149.000        -238.802

Difference of  238.802 in BIC' provides very strong support
for current model.
```

Diagnostics

Is not yet elaborated very well.

The multinomial logit implies a very special property: the independence of irrelevant alternatives (IIA). IIA means that the odds are independent from the other outcomes available (see the expression for P_j/P_0 above). IIA implies that estimates do not change, if the set of alternatives changes. This is a very strong assumption that in many settings will not hold. A general rule is that it holds, if outcomes are distinct. It does not hold, if outcomes are close substitutes.

There are different tests for this assumption. The intuitive idea is to compare the full model with a model, where one drops one outcome. If IIA holds, estimates should not change too much.

```
. mlogtest, iia

**** Hausman tests of IIA assumption
 Ho: Odds(Outcome-J vs Outcome-K) are independent of other alternatives.

 Omitted |      chi2   df   P>chi2   evidence
---------+-------------------------------------
     CDU |     0.486   15    1.000   for Ho
     SPD |    -0.351   14     ---    for Ho
     FDP |    -4.565   14     ---    for Ho
   Gruene|    -2.701   14     ---    for Ho
     PDS |     1.690   14    1.000   for Ho
-----------------------------------------------
 Note: If chi2<0, the estimated model does not
 meet asymptotic assumptions of the test.

**** Small-Hsiao tests of IIA assumption

 Ho: Odds(Outcome-J vs Outcome-K) are independent of other alternatives.

 Omitted |  lnL(full)  lnL(omit)    chi2   df   P>chi2   evidence
---------+-------------------------------------------------------
     CDU |   -903.280   -893.292  19.975    4    0.001   against Ho
     SPD |   -827.292   -817.900  18.784    4    0.001   against Ho
     FDP |  -1243.809  -1234.630  18.356    4    0.001   against Ho
   Gruene|  -1195.596  -1185.057  21.076    4    0.000   against Ho
     PDS |  -1445.794  -1433.012  25.565    4    0.000   against Ho
-------------------------------------------------------------------
```

In our case the results are quite inconclusive! The tests for the IIA assumption do not work well.

A related question with practical value is, whether we could simplify our model by collapsing categories:

```
. mlogtest, combine

**** Wald tests for combining outcome categories

 Ho: All coefficients except intercepts associated with given pair
     of outcomes are 0 (i.e., categories can be collapsed).

Categories tested |      chi2   df   P>chi2
------------------+-----------------------
     CDU-     SPD |    35.946    3    0.000
     CDU-     FDP |    33.200    3    0.000
     CDU-   Gruene|   156.706    3    0.000
     CDU-     PDS |    97.210    3    0.000
     CDU-   other |    52.767    3    0.000
     SPD-     FDP |     8.769    3    0.033
     SPD-   Gruene|   103.623    3    0.000
     SPD-     PDS |    79.543    3    0.000
     SPD-   other |    26.255    3    0.000
     FDP-   Gruene|    35.342    3    0.000
     FDP-     PDS |    61.198    3    0.000
     FDP-   other |    23.453    3    0.000
  Gruene-     PDS |    86.508    3    0.000
  Gruene-   other |    35.940    3    0.000
     PDS-   other |    88.428    3    0.000
-------------------------------------------
```

The parties seem to be distinct alternatives.

6) Models for Ordinal Outcomes

Models for ordinal dependent variables can be formulated as a threshold model with a latent dependent variable:

$$y^* = \beta' \mathbf{x} + \epsilon,$$

where Y^* is a latent opinion, value, etc. What we observe is

$$\begin{aligned} y &= 0, \quad \text{if} \quad y^* \leq \mu_0, \\ y &= 1, \quad \text{if} \quad \mu_0 < y^* \leq \mu_1, \\ y &= 2, \quad \text{if} \quad \mu_1 < y^* \leq \mu_2, \\ &\vdots \\ y &= J, \quad \text{if} \quad \mu_{J-1} < y^*. \end{aligned}$$

μ_j are unobserved thresholds (also termed cutpoints). We have to estimate them together with the regression coefficients. The model constant and the thresholds together are not identified. Stata restricts the constant to 0. Note that this model has only one coefficient vector.

One can make different assumptions on the error distribution. With a logistic distribution we obtain the ordered logit, with the standard normal we obtain the ordered probit. The formulas for the ordered probit are:

$$\begin{aligned} P(Y = 0) &= \Phi(\mu_0 - \beta' \mathbf{x}), \\ P(Y = 1) &= \Phi(\mu_1 - \beta' \mathbf{x}) - \Phi(\mu_0 - \beta' \mathbf{x}), \\ P(Y = 2) &= \Phi(\mu_2 - \beta' \mathbf{x}) - \Phi(\mu_1 - \beta' \mathbf{x}), \\ &\vdots \\ P(Y = J) &= 1 - \Phi(\mu_{J-1} - \beta' \mathbf{x}). \end{aligned}$$

For J=1 we obtain the binary probit. Estimation is done by ML.

Interpretation

We can use a sign interpretation on Y*. Very simple and often the only interpretation that we need.

To give more concrete interpretations one would want a

probability interpretation. The formula for the marginal effects is

$$\frac{\partial P(Y=j)}{\partial x_j} = (\phi(\mu_{j-1} - \beta'\mathbf{x}) - \phi(\mu_j - \beta'\mathbf{x}))\beta_j.$$

Again, they depend on x, there sign can be different from β, and even change as x changes.

Discrete probability effects are even more informative. One computes predicted probabilities and computes discrete effects. Predicted probabilities can be used to construct conditional-effect plots.

An example: Opinion on gender role change

Dependent variable is an item on gender role change (woman works, man keeps the house). Higher values indicate that the respondent does not dislike this change. The variable is named ”newrole”. It has 3 values. Independent variables are religiosity, woman, east. This is the result from an oprobit.

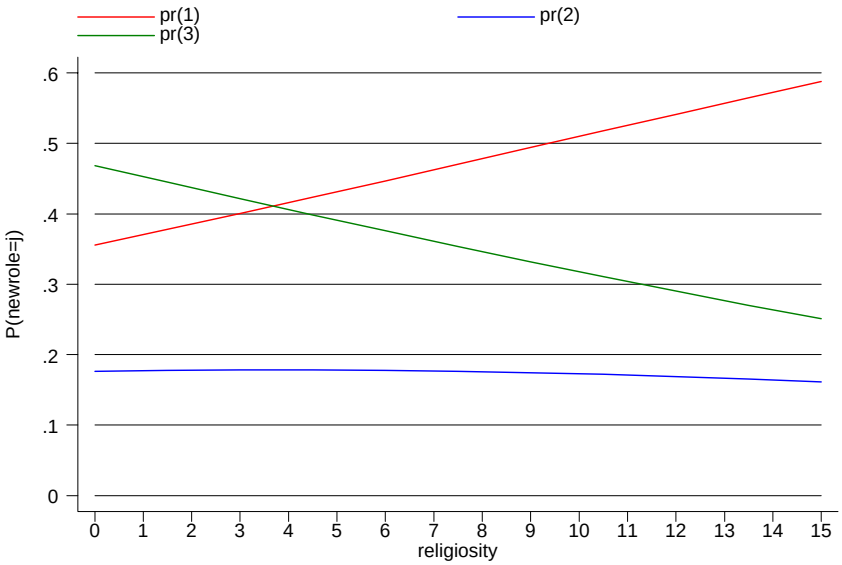
```
. oprobit newrole relig woman east, table

Iteration 0:   log likelihood = -3305.4263
Iteration 1:   log likelihood = -3256.7928
Iteration 2:   log likelihood = -3256.7837

Ordered probit estimates           Number of obs   =       3195
                                   LR chi2(3)      =      97.29
                                   Prob > chi2     =     0.0000
Log likelihood = -3256.7837        Pseudo R2       =     0.0147

------------------------------------------------------------
     newrole |      Coef.   Std. Err.      z    P>|z|
-------------+----------------------------------------------
       relig |  -.0395053   .0049219    -8.03   0.000
       woman |    .291559   .0423025     6.89   0.000
        east |  -.2233122   .0483766    -4.62   0.000
-------------+----------------------------------------------
       _cut1 |   -.370893    .041876     (Ancillary parameters)
       _cut2 |   .0792089   .0415854
------------------------------------------------------------

     newrole |      Probability            Observed
-------------+-------------------------------------
           1 |   Pr(      xb+u<_cut1)        0.3994
           2 |   Pr(_cut1<xb+u<_cut2)        0.1743
           3 |   Pr(_cut2<xb+u)              0.4263


. fitstat
```

```
Measures of Fit for oprobit of newrole

Log-Lik Intercept Only:     -3305.426  Log-Lik Full Model:    -3256.784
D(3190):                     6513.567  LR(3):                    97.285
                                       Prob > LR:                 0.000
McFadden's R2:                  0.015  McFadden's Adj R2:         0.013
Maximum Likelihood R2:          0.030  Cragg & Uhler's R2:        0.034
McKelvey and Zavoina's R2:      0.041
Variance of y*:                 1.042  Variance of error:         1.000
Count R2:                       0.484  Adj Count R2:              0.100
AIC:                            2.042  AIC*n:                  6523.567
BIC:                       -19227.635  BIC':                    -73.077
```

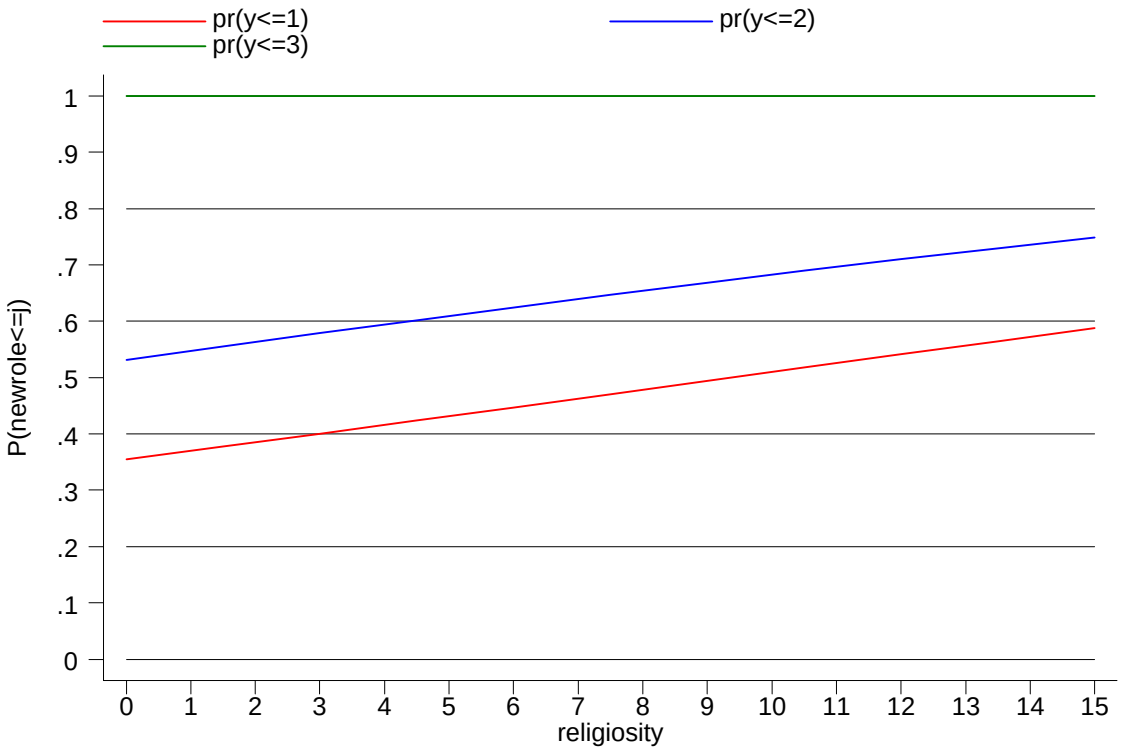
The fit is poor, which is common in opinion research.

```
. prchange

oprobit: Changes in Predicted Probabilities for newrole

relig
            Avg|Chg|           1           2           3
Min->Max   .15370076    .23055115  -.00770766  -.22284347
   -+1/2   .0103181     .01523566   .00024147  -.01547715
  -+sd/2   .04830311    .0713273    .00112738  -.07245466
MargEfct   .0309562     .01523658   .00024152  -.0154781

woman
            Avg|Chg|           1           2           3
    0->1   .07591579   -.1120384  -.00183527   .11387369

east
            Avg|Chg|           1           2           3
    0->1   .05785738   .08678606  -.00019442  -.08659166
```

Finally, we produce a conditional effect plot (man, west).



Even nicer is a plot of the cumulative predicted probabilities (especially if Y has many categories).




```
STATA syntax:
prgen relig, from(0) to(15) x(east 0 woman 0) gen(w)
gr7 wp1 wp2 wp3 wx, c(lll) s(iii) ylabel(0(.1).6) xlabel(0(1)15)
gr7 ws1 ws2 ws3 wx, c(lll) s(iii) ylabel(0(.1)1) xlabel(0(1)15)
```

The ordinal probit (logit) model includes a parallel regression assumption. The formulas above imply

$$P(Y \leq j|\mathbf{x}) = \Phi(\mu_j - \beta'\mathbf{x}).$$

This defines a set of binary response models with identical slope. We could run binary probits on the outcomes defined as 1 if $y \leq j$, 0 else. The probit coefficients should be equal. There are formal tests for this assumption.

```
. omodel probit newrole relig woman east

Approximate likelihood-ratio test of equality of coefficients
across response categories:
         chi2(3) =     12.18
       Prob > chi2 =    0.0068
```

In our example the parallel regression assumption is violated. An alternative would be the multinomial model.

7) Models for Special Data Situations

In this chapter we will discuss several (cross-sectional) regression models for special kinds of data.

Models for count data

With count data $Y \in \{0, 1, 2, 3, \ldots\}$. Count data can be seen as the result of an event generating recurrent process. They then count the number of events. If the event rate is a constant μ the number of events (counts) follows a Poisson distribution (for a fixed exposure interval of 1)

$$P(Y = y) = \frac{e^{-\mu}\mu^y}{y!}, \qquad y = 0, 1, 2, \ldots$$

where $\mu > 0$. It is $E(y) = V(y) = \mu$. The mean and variance are identical. This property of the Poisson distribution is known as equidispersion. We obtain a regression model by specifying

$$\mu_i = E(y_i|\mathbf{x}_i) = \exp(\beta'\mathbf{x}_i).$$

This is the *Poisson regression model*. e^β gives the discrete (multiplicative) effect on the expected count. The absolute effect on μ can be calculated either as marginal or discrete effect. Both depend on the value of X.

Often data will show over-dispersion (V>E: event rate increases, infection, ...) or under-dispersion (V<E; event rate declines). With over-dispersed data you could use the *negative binomial regression model*. This model adds unobserved heterogeneity by specifying $\mu_i = \exp(\beta'\mathbf{x}_i)\epsilon_i$. One assumes that ϵ_i follows a gamma distribution. This sounds nice, but it is nevertheless a very strong assumption. Therefore, be careful when using such models.

Finally, there is a class of models called ”*zero inflated*” count models. These models assume that there are two latent classes of observations: those who can only have a 0 count (probability 1 for a zero), and those who have a positive probability for any count. In many applications this makes sense. In the example below, for instance, some woman can have no children for

biological reasons. Whether people belong to first or second group is modeled by a logit. The count model for the second group is modeled either as Poisson or negative binomial. These models assume also over-dispersion.

An Example using Stata

Using ALLBUS 1982, 1984, and 1991 we analyze the number of children of German woman over 39. We use only woman born after 1929 (they were at ”risk” during the existence of FRG and GDR), and who were born and interviewed in West/East.

The restriction to woman over 39 is used to have an identical exposure time. Otherwise we would have to include an offset t (t=time at risk).

As independent variables we consider birth cohort (30/34=1, 35/39=2, 40/44=3, 45/52=4), whether the woman was ever married, education, and West/East.

First, we run an OLS regression:

```
. regr    nchild coh2 coh3 coh4 marr educ east

R-squared     =  0.1217

------------------------------------------------------
      nchild |      Coef.   Std. Err.      t    P>|t|
-------------+----------------------------------------
        coh2 |  -.1305614   .0752871    -1.73   0.083
        coh3 |  -.3584656   .0790622    -4.53   0.000
        coh4 |   -.382933   .0852924    -4.49   0.000
        marr |   1.785363   .1267655    14.08   0.000
        educ |  -.0187562   .0180205    -1.04   0.298
        east |   .1369749   .0611933     2.24   0.025
       _cons |   .6022025   .2175236     2.77   0.006
------------------------------------------------------
```

Now a Poisson regression (IRR=incidence rate ratio=e^{β}):

```
. poisson nchild coh2 coh3 coh4 marr educ east, irr

Poisson regression                  Number of obs   =       1805
                                    LR chi2(6)      =     262.83
                                    Prob > chi2     =     0.0000
Log likelihood = -2782.6208         Pseudo R2       =     0.0451
```

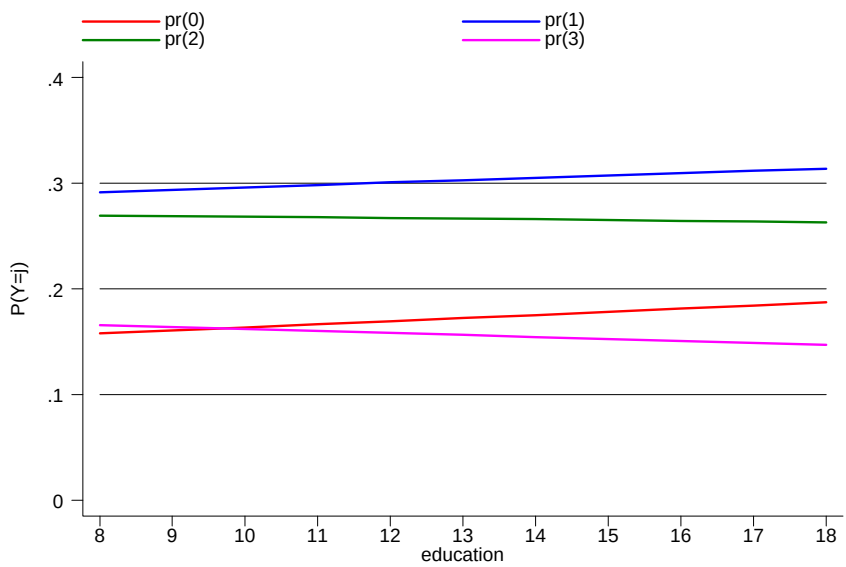
```
------------------------------------------------------------
      nchild |        IRR   Std. Err.      z    P>|z|
-------------+----------------------------------------------
        coh2 |   .9408361   .0413808    -1.39   0.166
        coh3 |   .8339971   .0400824    -3.78   0.000
        coh4 |   .8246442    .042896    -3.71   0.000
        marr |   8.814683   1.931314     9.93   0.000
        educ |   .9902484    .011152    -0.87   0.384
        east |   1.072145   .0394467     1.89   0.058
------------------------------------------------------------
```

Note that there are some differences (East is now insignificant). Now we compute effects on μ.

```
. prchange

poisson: Changes in Predicted Rate for nchild

      min->max      0->1     -+1/2    -+sd/2  MargEfct
coh2   -0.1102   -0.1102   -0.1116   -0.0513   -0.1116
coh3   -0.3179   -0.3179   -0.3325   -0.1442   -0.3320
coh4   -0.3335   -0.3335   -0.3532   -0.1416   -0.3527
marr    1.8119    1.8119    4.8146    0.8842    3.9810
educ   -0.0884   -0.0195   -0.0179   -0.0284   -0.0179
east    0.1293    0.1293    0.1274    0.0583    0.1274

exp(xb):   1.8292

        coh2     coh3     coh4     marr     educ     east
 x=  .303601  .252078  .201662   .94903  9.03324  .297507
sd(x)=.45994  .434326  .401352  .219996  1.58394  .457288
```

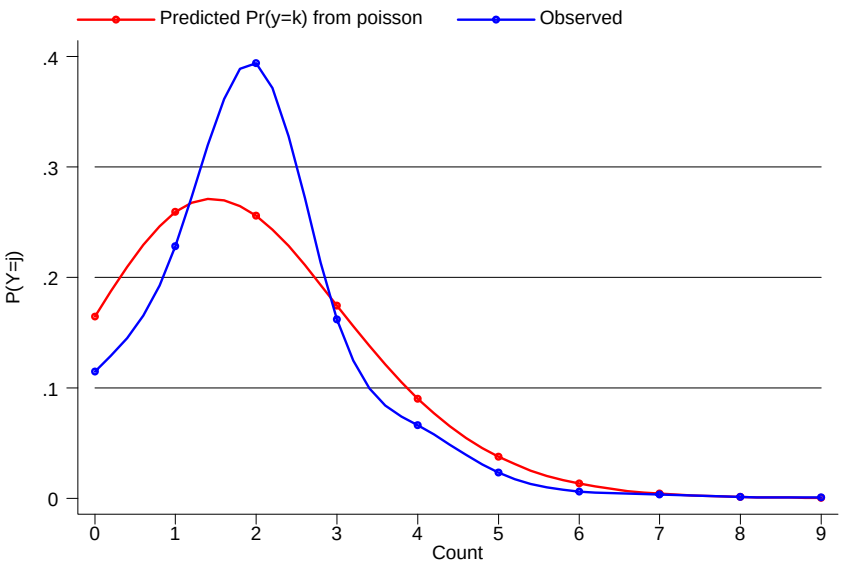
Note that the centered and marginal effects of marr are nonsense! Better we plot conditional effect plots.

```
prgen educ, from(8) to(18) rest(mean) gen(pr)
gr7 prp0 prp1 prp2 prp3 prx, c(llll) s(iiii)
```



The fit of the Poisson can be assessed by comparing observed and predicted probabilities.

```
prcounts w, plot
gr7 wpreq wobeq wval, c(ss) s(oo)
```



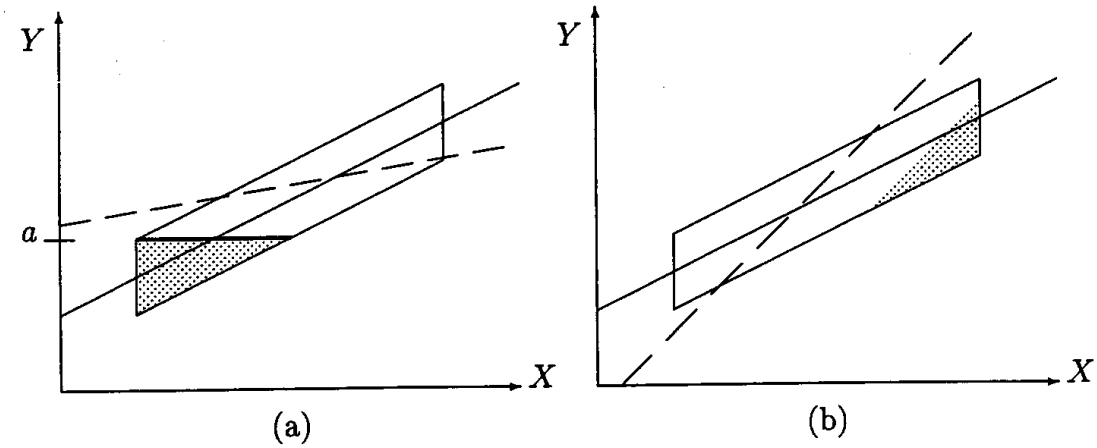
The fit is quite bad. So we try the negative binomial.

```
. nbreg   nchild coh2 coh3 coh4 marr educ east

Fitting full model:

Iteration 0:   log likelihood = -2791.3516
Iteration 1:   log likelihood = -2782.6306
Iteration 2:   log likelihood = -2782.6208
Iteration 3:   log likelihood = -2782.6208  (not concave)

Negative binomial regression                      Number of obs   =
......
-------------+----------------------------------------------------------------
       alpha |   2.13e-11          .                              .
------------------------------------------------------------------------------
```

It does not work, because our data are *under-dispersed* (E=1.96, V=1.57). For the same reason the zero inflated models also do not work.

Censored and truncated data

Censoring occurs, when some observations on the dependent variable report not the true value but a cutpoint. Truncation means that complete observations beyond a cutpoint are missing. OLS estimates with censored or truncated data are biased.



In (a) data are censored at *a*. One knows that there true value is *a* or less. The regression line would be less steep (dashed line). Truncation means that cases below *a* are completely missing. Truncation also biases OLS estimates. (b) is the case of incidential truncation or sample selection. Due to a non-random selection mechanism information on Y is missing for some cases. This biases OLS estimates also. Therefore, special estimation methods exist for such data.

Censored data are analyzed with the *tobit model* (s. Long: ch. 7):

$$y_i^* = \beta' \mathbf{x}_i + \epsilon_i,$$

where $\epsilon_i \sim N(0, \sigma^2)$. Y^* is the latent uncensored dependent variable. What we observe is

$$y_i = 0, \quad \text{if} \quad y_i^* \leq 0,$$
$$y_i = y_i^*, \quad \text{if} \quad y_i^* > 0.$$

Estimation is done by ML (analogous to event history models!).

β is a discrete effect on the latent, uncensored variable

$$\frac{\partial E(y^*|\mathbf{x})}{\partial x_j} = \beta_j.$$

This interpretation makes sense, because the scale of Y^* is known. Interpretation in terms of Y is more complicated. One has to multiply coefficients by a scale factor

$$\frac{\partial E(y|\mathbf{x})}{\partial x_j} = \beta_j \Phi\left(\frac{\beta'\mathbf{x}}{\sigma}\right).$$

Example: Income artificially censored

I censor ”income” (ALLBUS 1994) at 10,001.- DM. 12 observations are censored. I used the following to compare OLS with the original data (1), OLS with the censored data (2), and tobit (3).

```
regress income   educ exp prestf woman east white civil self
outreg using tobit, replace
regress incomec  educ exp prestf woman east white civil self
outreg using tobit, append
tobit incomec  educ exp prestf woman east white civil self, ul
outreg using tobit, append
```

	(1) income	(2) incomec	(3) incomec
educ	182.904	179.756	180.040
	(10.48)**	(11.88)**	(11.84)**
exp	26.720	25.947	25.981
	(7.28)**	(8.16)**	(8.12)**
prestf	4.163	3.329	3.356
	(2.92)**	(2.70)**	(2.71)**
woman	-797.766 -	785.853	-786.511
	(8.62)**	(9.80)**	(9.76)**
east	-1,059.817	-1,032.873	-1,034.475
	(12.21)**	(13.73)**	(13.68)**
white	379.924	391.658	391.203
	(3.71)**	(4.41)**	(4.38)**
civil	419.790	452.013	450.250
	(2.43)*	(3.02)**	(2.99)**
self	1,163.615	925.104	941.097
	(8.10)**	(7.43)**	(7.52)**
Constant	52.905	131.451	127.012
	(0.24)	(0.70)	(0.67)
R-squared	0.34	0.38	

Absolute value of t statistics in parentheses
* significant at 5%; ** significant at 1%

OLS estimates in (2) are biased. The tobit improves only a little on this. This is due to the nonnormality of our dependent variable. The whole tobit procedure rests essentially on the

assumption of normality. If it is not fullfilled, it does not work. This shows that sophisticated econometric methods are not robust. So why not use OLS?

Regression Models for Complex Survey Designs

Most estimators and its standard errors are derived under the assumption of simple random sampling with replacement (SRSWR). However in practice many surveys involve more complex sampling schemes:

- the sampling probabilities might differ between the observations
- the observations are sampled randomly within clusters (PSU's)
- the observations are drawn independently from different stratas.

The ALLBUS 94 samples respondents within constituencies. In other words, a twostage sampling is used. If we use estimators that assume independence, the standard errors may be too small. However Statas `svy`-commands are able to correct the standard errors for many estimation commands. Therefore you need to declare your data to be "svy"-data and estimate the appropriate svy-regression model:

```
. svyset, psu(v350)   /* We use the intnr as primary sampling unit */
. svyreg eink bild exp prest frau ost angest beamt selbst, deff

Survey linear regression
pweight:  <none>              Number of obs    =      1240
Strata:   <one>               Number of strata =         1
PSU:      v350                Number of PSUs   =       486
                              Population size  =      1240
                              F(   8,    478)  =     78.02
                              Prob > F         =    0.0000
                              R-squared        =    0.3381
------------------------------------------------------
        eink |      Coef.   Std. Err.      Deff
-------------+----------------------------------------
        bild |   182.9042   21.07473   1.079241
         exp |   26.71962   3.411434   1.031879
       prest |   4.163393   1.646775   .9809116
        frau |  -797.7655   86.53358   .9856359
         ost |  -1059.817    75.4156   1.091877
      angest |   379.9241   84.19078   1.001129
       beamt |   419.7903   128.1363   1.126659
      selbst |   1163.615   273.5306   1.064807
```

```
       _cons |     52.905    255.014   1.096803
-----------------------------------------------------
```

The point estimates are equal to the point estimates of the simple OLS-Regression. But the standard errors differ. Kishs Designeffekt deff shows the multiplicative difference between the ”true” standard error and the standard error of the simple regression model.

Note that the svy-estimators allows any level of correlation within the primary sampling unit. Thus elements within a primary sampling unit do not have to be independent. There can be a secondary clustering.

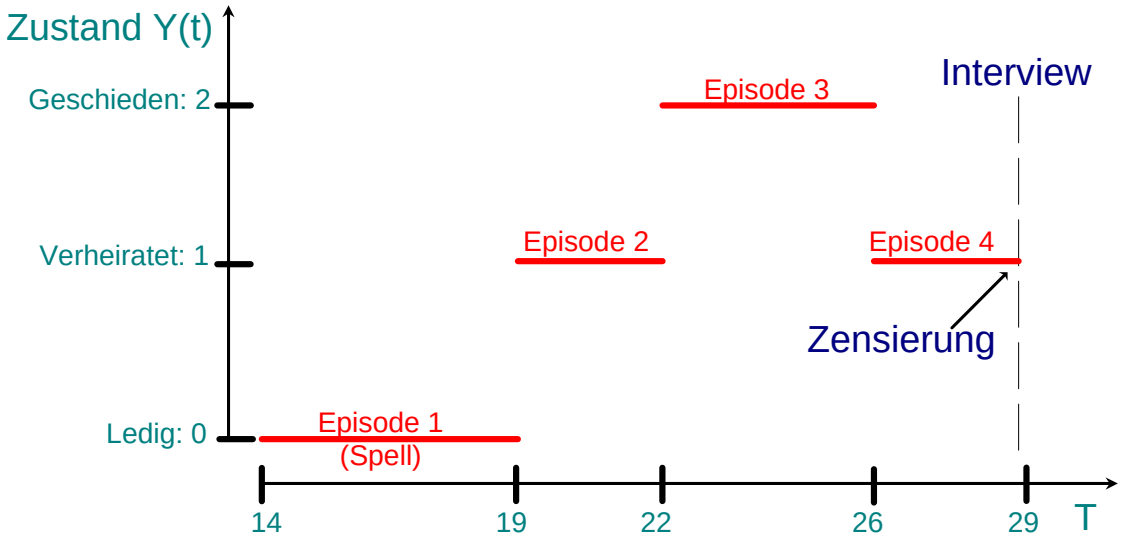
In many surveys, observations have different probabilities of selection. Therefore one needs a weighting variable which is equal (or proportional) to the inverse of the probability beeing sampled. If we omit the weights in the analysis, the estimates may be (very) biased. Weights also affect the standard errors of the estimates. To include weights in the analysis we can use another `svyset` command. Below you find an example with houshold size for illustration.

```
. svyset [pweight = v266]
. svyreg eink bild exp prest frau ost angest beamt selbst, deff

Survey linear regression
pweight:  v266                      Number of obs    =      1240
Strata:   <one>                     Number of strata =         1
PSU:      v350                      Number of PSUs   =       486
                                    Population size  =      3670
                                    F(   8,    478)  =     58.18
                                    Prob > F         =    0.0000
                                    R-squared        =    0.3346
-------------------------------------------------------------
        eink |      Coef.   Std. Err.      Deff
-------------+-----------------------------------------------
        bild |   180.6797   24.43859   1.389275
         exp |    29.8775   4.052303   1.204561
       prest |   5.164107   2.197095   1.351514
        frau |  -895.3112   102.0526   1.186356
         ost |  -1084.513   85.35748   1.395625
      angest |   441.0447   101.0716     1.2316
       beamt |   437.3239   145.5182   1.284389
      selbst |    1070.29   300.7471   1.408905
       _cons |   35.99856   308.3018   1.426952
-------------------------------------------------------------
```

8) Event History Models

Longitudinal data add a time dimension. This makes it easier to identify ”causal” effects, because one knows the time ordering of the variables. Longitudinal data come in two kinds: event history data or panel data.

Event history data record the life course of persons.



marital ”career” of a person

Event history data record the age something happens and the state afterwards:

$$(14,0)(19,1)(22,2)(26,1)(29,1).$$

From this we can compute the duration until an event happens: t=5 for first marriage, t=3 for divorce, t=4 for second marriage, t=3 for second divorce (this duration, however, is censored!). These durations are the dependent variable in event history regressions.

For this example taking regard if the time ordering could mean that we look for the effects of career history on later events. Or we could measure parallel careers. For instance we could investigate how events from the labor market career affect the marital career.

The accelerated failure time model

We model the duration (T) until an event takes place by:

$$\ln t_i = \boldsymbol{\beta}_*' \mathbf{x}_i + \epsilon_i.$$

This is the accelerated failure time model. Depending on the distribution of the error term that we assume, different regression models result. If we assume the logistic we get the log-logistic regression model. Other models are: exponential, Weibull, lognormal, gamma. e^{β_*} gives the (multiplicative) discrete unit effect on the time scale (the factor by which time is accelerated, or decelerated).

Some basic concepts

However, this is not the standard specification for event history regression models. Usually, one uses an equivalent specification in terms of the (hazard) rate function. Thus, we first need to discuss this concept.

A rate is defined as:

$$r(t) = \lim_{\Delta t \to 0} \frac{P(t + \Delta t > T > t | T > t)}{\Delta t}.$$

It gives approximately the conditional probability of having an event, given that one did not have an event up to t. A rate function describes the distribution of T.

An alternative way to define it is by

$$r(t) = \frac{f(t)}{S(t)},$$

where f(t) is the density function and S(t) is the survival function. f(t) is the (unconditional) probability to have an event at t. S(t) gives the proportion that did not have an event up to t.

From this one can derive

$$S(t) = e^{-\int_0^t r(u)du}.$$

Proportional hazard regression model

This is most widely used specification of a rate regression. We assume that X has a proportional effect on the rate. We model conditional rate functions as

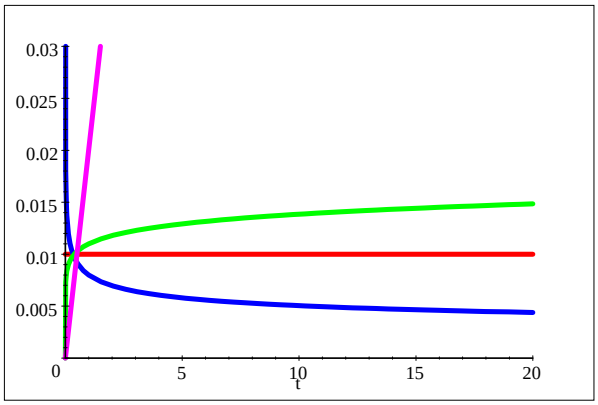
$$r(t|x) = r_0(t)e^{\beta x} = r_0(t) \cdot \alpha^x.$$

$r_0(t)$ is a base rate, $e^\beta = \alpha$ is the (multiplicative) discrete effect on the rate (termed ”relative risk”). $(\alpha - 1)100$ is a percentage effect (compare with semi-logarithmic regression).

To complete the specification one has to specify a base rate:

Exponential model (constant rate model): $r_0(t) = \alpha_0$.

Weibull model (p is a shape parameter): $r_0(t) = p\,t^{p-1}\alpha_0$.



$\alpha_0 = 0.01$

blue: p=0.8

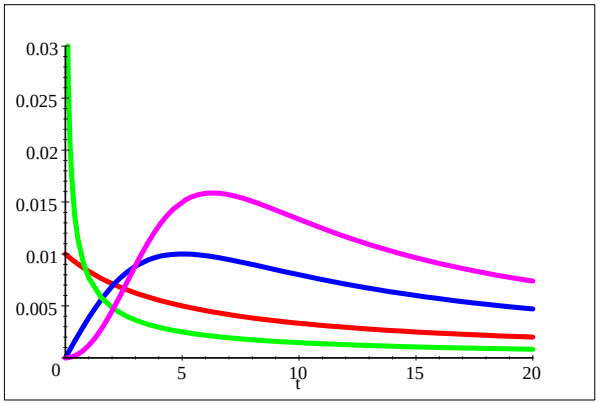
red: p=1

green: p=1.1

violet: p=2

Generalized log-logistic model: (p: shape, λ: scale)

$$r_0(t) = \frac{p(\lambda t)^{p-1}}{1 + (\lambda t)^p}\alpha_0.$$



$\alpha_0 = 0.01,\ \lambda = 0.2$

green: p=0.5

red: p=1

blue: p=2

violet: p=3

ML estimation

One has to take regard of the censored durations. It would bias results, if we would drop these. This is, because censored durations are informative: The respondent did not have an event until t. To indicate which observation ends by an event, and which one is censored we define an censoring indicator Z: z=1 for durations ending by an event, z=0 for censored durations. The we can formulate the likelihood function:

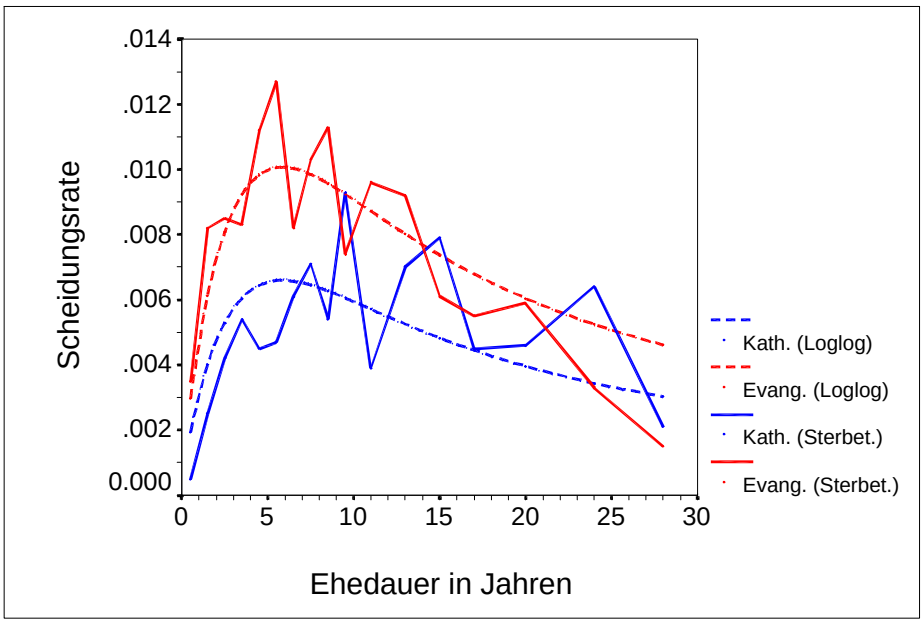
$$L(\alpha) = \prod_{i=1}^{n} f(t_i;\alpha)^{z_i} \cdot S(t_i;\alpha)^{1-z_i} = \prod_{i=1}^{n} r(t_i;\alpha)^{z_i} \cdot S(t_i;\alpha).$$

The log likelihood is

$$\ln L(\alpha) = \sum_{i=1}^{n} \left[z_i \cdot \ln r(t_i;\alpha) - \int_0^{t_i} r(u;\alpha)du \right].$$

Example: Divorce by religion

Data are from the German Family Survey 1988. We model duration of first marriage by religion (0=protestant, 1=catholic). Solid lines are non parametric rate estimators (life-table), dashed lines are estimates from the generalized log-logistic.



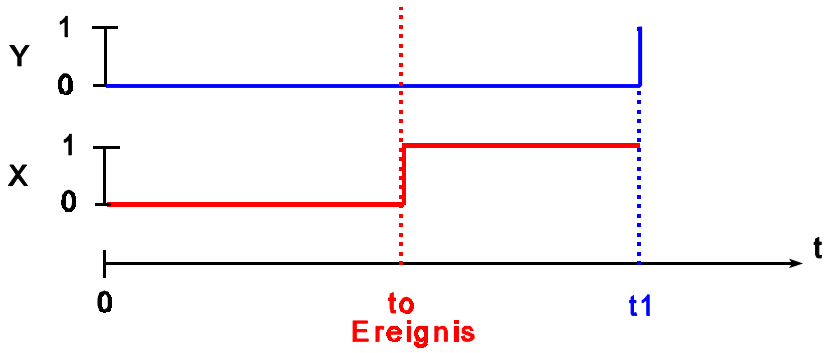
The model fits the data quite well. $\alpha = 0.65$, i.e. relative divorce risk is lower by the factor 0.65 for catholics (-35%).

Cox regression

To avoid a parametric assumption concerning the base rate, the Cox model does not specify it. Then, however, one cannot use ML. Instead, one uses a partial-likelihood method. Note, that this model still assumes proportional hazards. This is the reason, why this model is often named a semi-parametric model.

This model is used very often, because one does not need to think about which rate model to use. But it gives no estimate of the base rate. If one has substantial interest in the pattern of the rate (as is often the case), one has to use a parametric model.

Further, with the Cox model it is easy to include *time-varying covariates*. These are variables that can change their values over time. The effects of such variables account for the time ordering of events. Thus, with time-varying covariates it is possible to investigate the effects of earlier event on later events! This is a very distinct feature of event history analysis.



Example: Cox regression on divorce rate

Data as above. We investigate whether the event ”birth of a child” has an effect on the event ”divorce”.

	β-effect	S.E.	z-value	relative risk (α)
cohort 61-70	0.58	0.15	3.89	1.78
cohort 71-80	0.86	0.16	5.22	2.36
cohort 81-88	0.87	0.26	3.37	2.39
age at marriage woman	-0.12	0.02	6.39	0.89
education man	-0.11	0.05	2.40	0.89
education woman	0.07	0.05	1.31	1.07
catholic	-0.40	0.10	3.87	0.67
cohabitation	0.62	0.13	4.92	1.85
birth of child (time-vary.)	-0.79	0.11	7.36	0.45
Pseudo-R^2	3.1%			

reference: marriage cohort 49-60, protestant, no cohab, no child.

An example using Stata

With the ALLBUS 2000 we investigate the fertility rate of West-German woman. Independent variables are education, prestige father, West/East, and marriage cohort (04/25=1, 26/40=2, 41/50=3, 51/65=4, 66/81=5)

First, we have to construct the ”duration” variable: age at birth of first child-14 for observations with a child, age at interview-14 for censored observations. Second, we need a censoring indicator: ”child” (1 if child, 0 else). Now we must ”stset” the data:

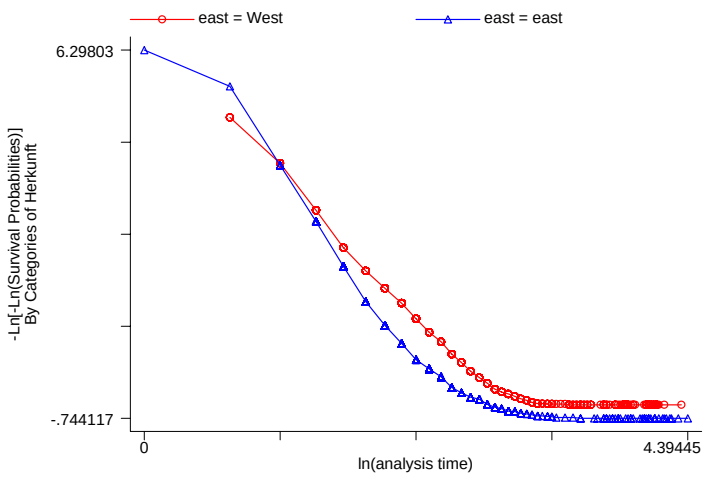
```
. stset duration, failure (child==1)

     failure event:  child == 1
obs. time interval:  (0, duration]
 exit on or before:  failure

------------------------------------------------------------------
     1472  total obs.
        0  exclusions
------------------------------------------------------------------
     1472  obs. remaining, representing
     1099  failures in single record/single failure data
    21206  total analysis time at risk, at risk from t =         0
                             earliest observed entry t =         0
                                  last observed exit t =        81
```

Next we run a Cox regression.

```
. stcox educ coh2 coh3 coh4 coh5 prestf east

         failure _d:  child == 1
   analysis time _t:  duration

Iteration 0:   log likelihood = -4784.5356
Iteration 1:   log likelihood = -4730.2422
Iteration 2:   log likelihood = -4729.6552
Iteration 3:   log likelihood =  -4729.655
Refining estimates:
Iteration 0:   log likelihood =  -4729.655

Cox regression -- Breslow method for ties

No. of subjects =         1043      Number of obs   =      1043
No. of failures =          761
Time at risk    =        14598
                                    LR chi2(7)      =    109.76
Log likelihood  =    -4729.655      Prob > chi2     =    0.0000

------------------------------------------------------------
          _t |
          _d | Haz. Ratio   Std. Err.      z    P>|z|
-------------+----------------------------------------------
        educ |   .9318186   .0159225    -4.13   0.000
        coh2 |   1.325748   .1910125     1.96   0.050
        coh3 |   1.773546   .2616766     3.88   0.000
        coh4 |   1.724948   .2360363     3.98   0.000
        coh5 |    1.01471   .1643854     0.09   0.928
      prestf |   .9972239   .0014439    -1.92   0.055
        east |   1.538249   .1147463     5.77   0.000
------------------------------------------------------------
```

We should test the proportionality assumption. Stata provides several methods to do this. We use a log-log plot of the survival functions. We test the variable West/East. The lines in this plot should be parallel.

```
. stphplot, by(east)
```



An disadvantage of the Cox model is that it provides no

information on the base rate. For this one could use a parametric regression model. Informal tests showed that a log-logistic rate model fits the data well.

```
. streg educ coh2 coh3 coh4 coh5 prestf east, dist (loglogistic)

Log-logistic regression -- accelerated failure-time form

No. of subjects =         1043        Number of obs   =      1043
No. of failures =          761
Time at risk    =        14598
                                      LR chi2(7)      =    146.49
Log likelihood  =   -996.50288        Prob > chi2     =    0.0000

------------------------------------------------------------
          _t |      Coef.   Std. Err.      z    P>|z|
-------------+----------------------------------------------
        educ |    .059984   .0095747     6.26   0.000
        coh2 |  -.2575441   .0892573    -2.89   0.004
        coh3 |  -.4696605   .0918465    -5.11   0.000
        coh4 |  -.4328219   .0845234    -5.12   0.000
        coh5 |  -.1753024    .091234    -1.92   0.055
      prestf |   .0017873   .0008086     2.21   0.027
        east |  -.3053707   .0426655    -7.16   0.000
       _cons |     2.1232    .117436    18.08   0.000
-------------+----------------------------------------------
     /ln_gam |  -.9669473   .0308627   -31.33   0.000
-------------+----------------------------------------------
       gamma |    .380242   .0117353
------------------------------------------------------------
```

Note that the log-logistic estimates the model with $\ln t$ as dependent variable. The coefficients are therefore β_*. The signs are therefore the opposite of the Cox model. Besides of this results are comparable. γ is the shape parameter (in the rate formulation it is 1/p). It indicates a non monotonic rate.

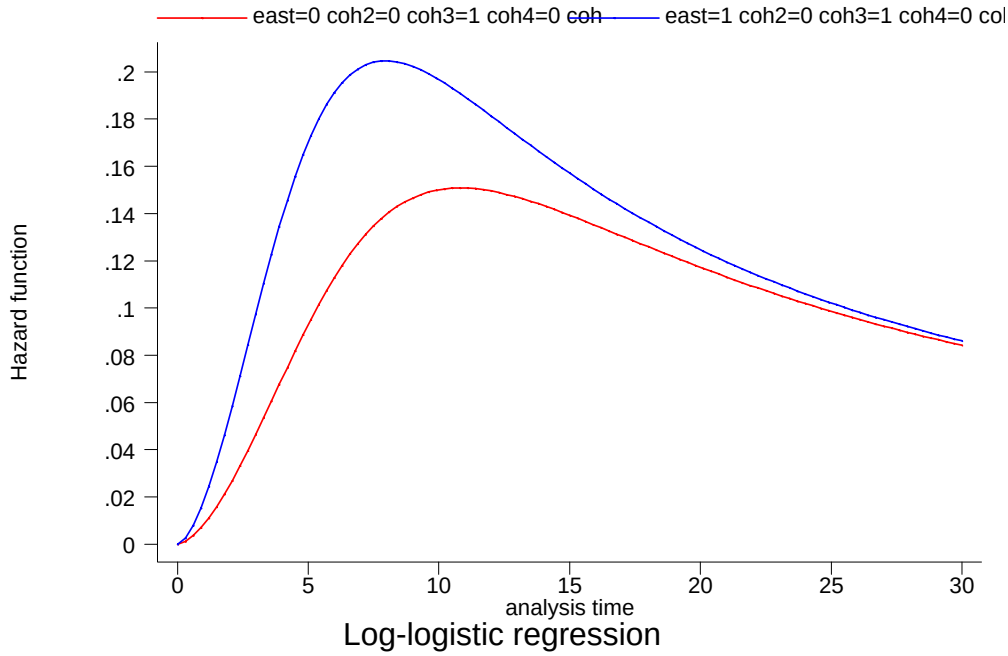
The magnitudes of these effects are not directly interpretable, but Stata offers some nice tools.

```
. streg, tr
```

produces e^{β_*}, the factor by which the time scale is multiplied (time ratios). But this is not very helpful.

A conditional rate plot:

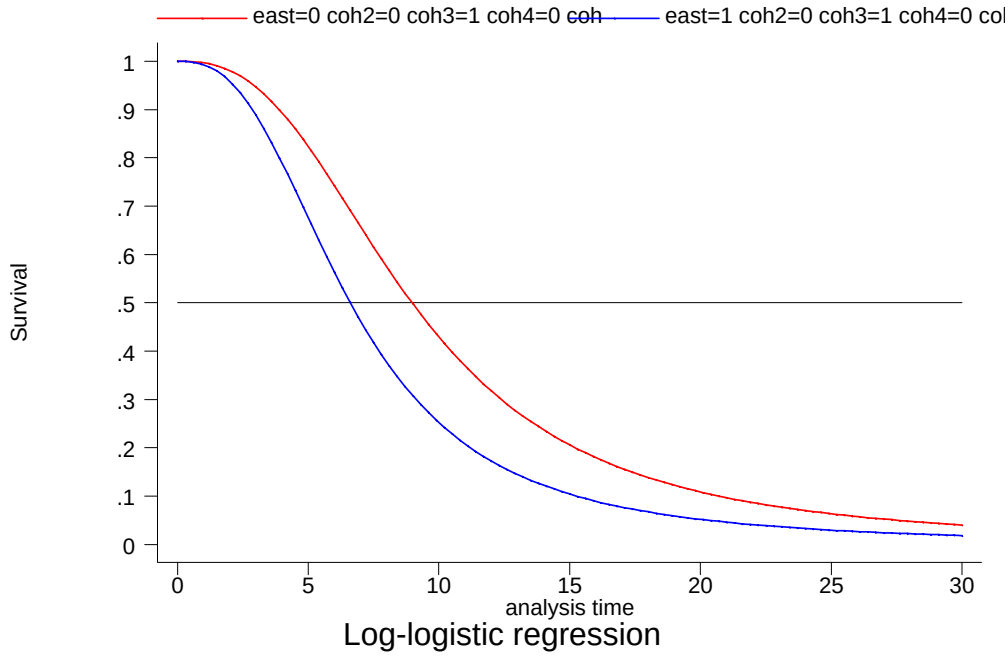
```
stcurve, hazard c(ll) s(..)
          at1(east=0 coh2=0 coh3=1 coh4=0 coh5=0 educ=9 prestf=0.5)
          at2(east=1 coh2=0 coh3=1 coh4=0 coh5=0 educ=9 prestf=0.5)
          ylabel(0(0.02)0.20) range(0 30) xlabel(0(5)30)
```



Note that the effect is not proportional!

A conditional survival plot:

```
stcurve, survival c(ll) s(..)
     at1(east=0 coh2=0 coh3=1 coh4=0 coh5=0 educ=9 prestf=0.5)
     at2(east=1 coh2=0 coh3=1 coh4=0 coh5=0 educ=9 prestf=0.5)
     ylabel(0(0.1)1) range(0 30) xlabel(0(5)30) yline(0.5)
```



Finally, we compute marginal effects on the median duration:

```
. mfx compute, predict(median time) nose

Marginal effects after llogistic
      y  = predicted median _t (predict, median time)
         =  12.289495
------------------------------------------------------------
       variable |          dy/dx                 X
----------------+-------------------------------------------
           educ |        .7371734            12.0086
           coh2*|       -2.916459            .171620
           coh3*|       -4.936661            .147651
           coh4*|       -5.017442            .347076
           coh5*|       -2.064034            .248322
         prestf |        .0219647            55.3915
           east |       -3.752852            .414190
-     ----------------------------------------------------------
(*) dy/dx is for discrete change of dummy variable from 0 to 1
```

A final remark for the experts: A next step would be to include time-varying covariates, e.g. marriage. For this, one would have to split the data set (using ”stsplit”).